

CSE 331: Microprocessor Interfacing and Embedded Systems

Lecture 4: ARM ISA (Assembly Language)

Books

- general reference :
 - W. Hohl, *ARM Assembly Language*, CRC Press, 2009
 - ARM7TDMI
 - NB not Linux format
- for Raspberry Pi:
 - B. Smith, *Raspberry Pi Assembly Language Raspbian Hands On Guide (2nd Ed)*, 2014

References

ARM Cortex A processor:

https://silver.arm.com/download/Software/BX100-DA-98001-r0p0-01rel3/DEN0013D_cortex_a_series_PG.pdf

General Purpose IO:

<https://www.raspberrypi.org/documentation/hardware/raspberrypi/bcm2835/BCM2835-ARM-Peripherals.pdf>

Memory

Memory is arranged as a series of “locations”

Each location has a unique “address”

Each location holds a byte (**byte-addressable**)

e.g. the memory location at address

0x080001B0 contains the byte value 0x70,
i.e., 112

The number of locations in memory is limited

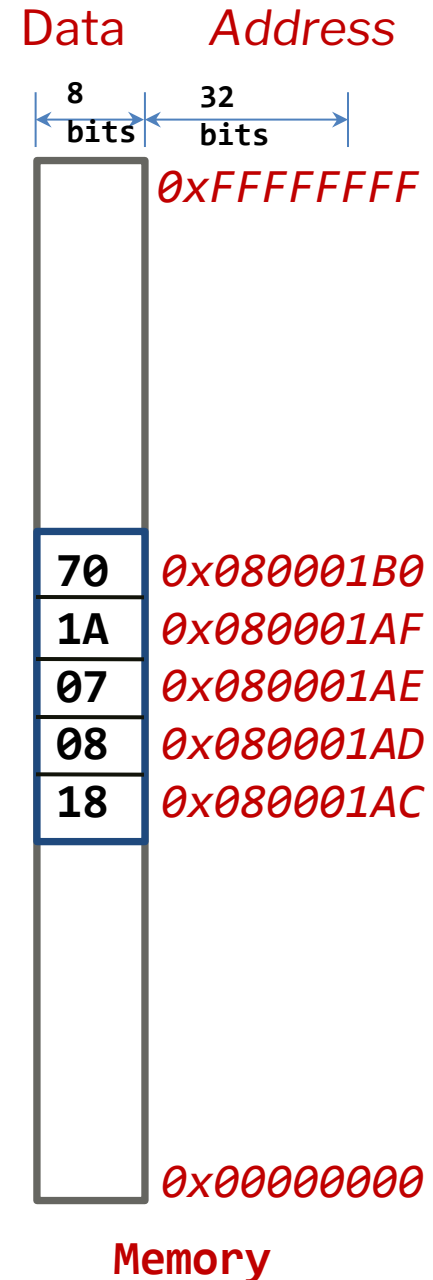
e.g. 4 GB of RAM

1 Gigabyte (GB) = 2^{30} bytes

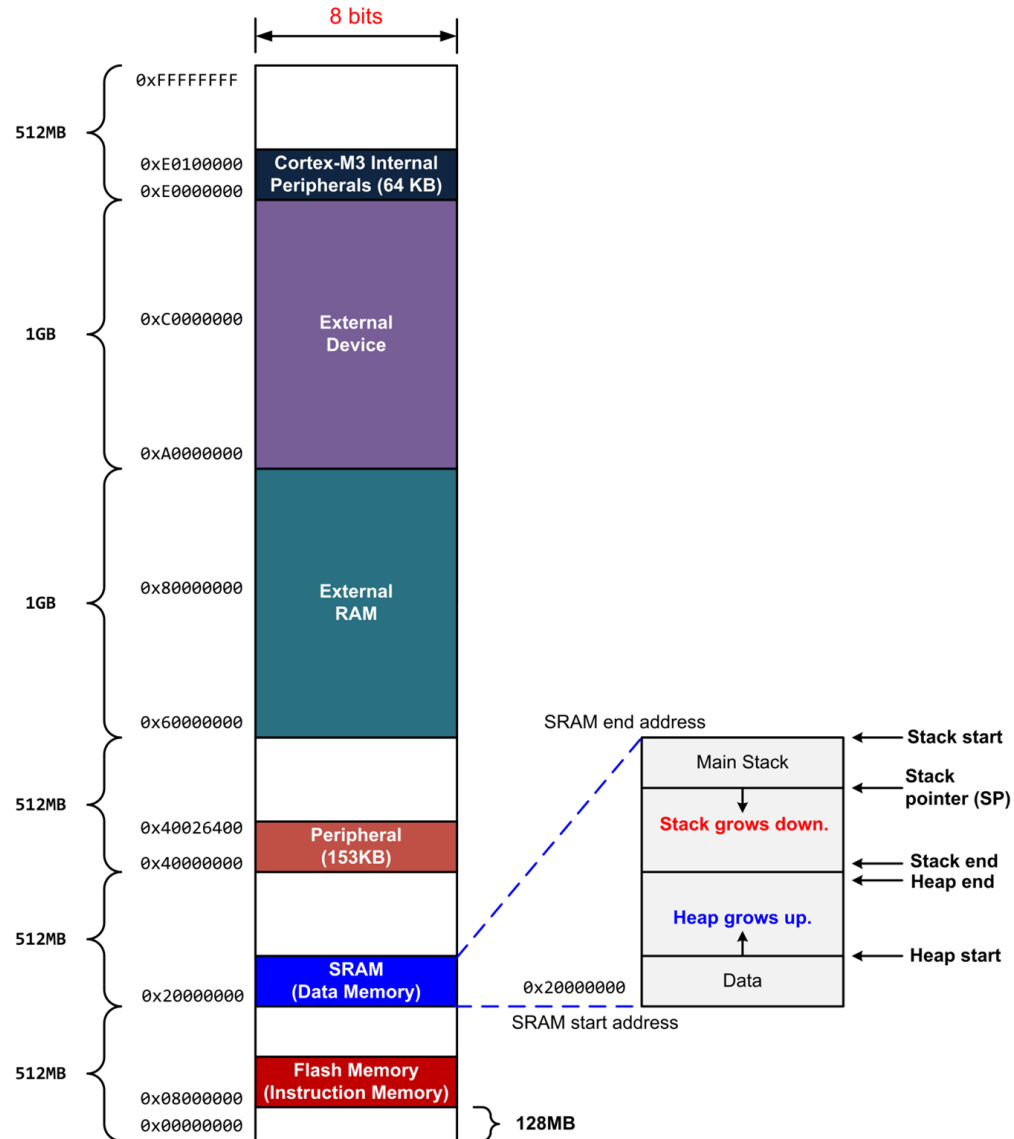
2^{32} locations = 4,294,967,296 locations!

Values stored at each location can represent either
program data or **program instructions**

e.g. the value 0x70 might be the code used to
tell the processor to add two values together



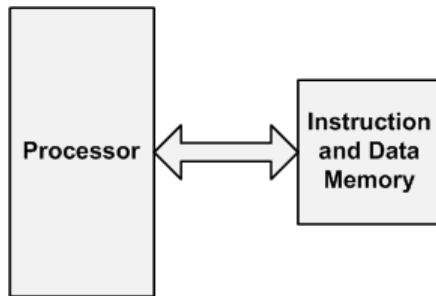
ARM M4 Memory Map



Computer Architecture

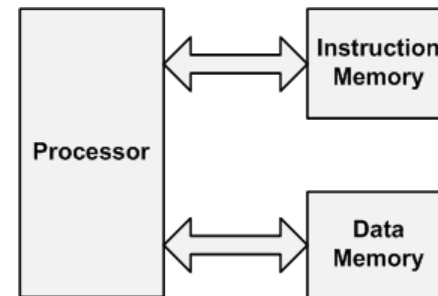
Von-Neumann

Instructions and data are stored in the same memory.



Harvard

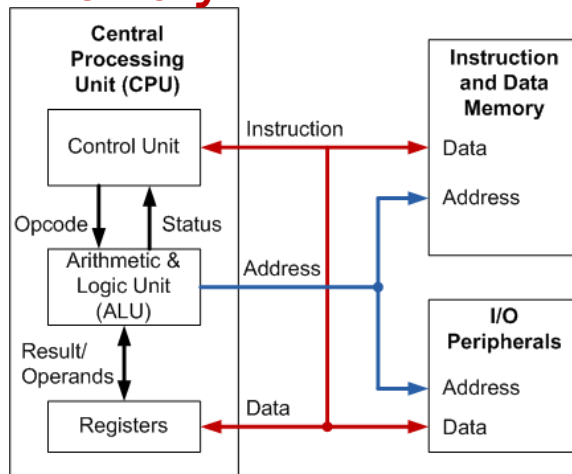
Data and instructions are stored into separate memories.



Computer Architecture

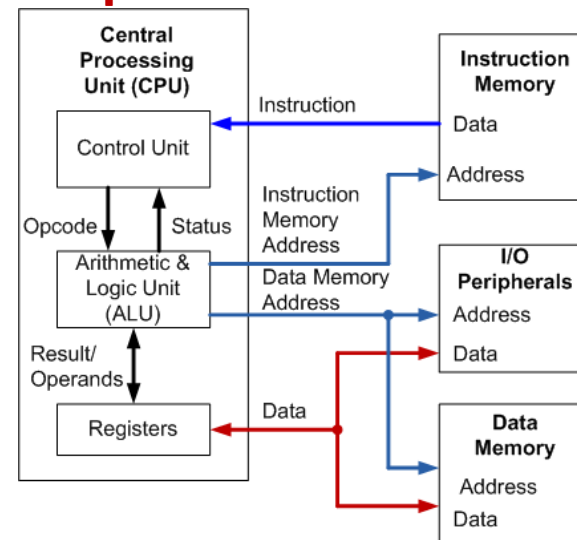
Von-Neumann

Instructions and data are stored in the same memory.



Harvard

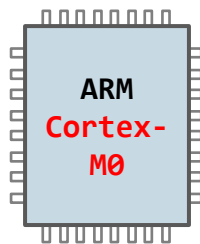
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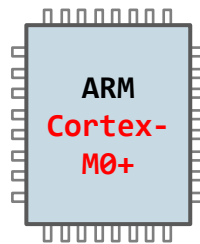
ARM Cortex-M Series Family

Von-Neumann

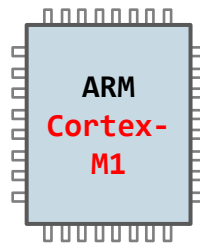
Instructions and data are stored in the same memory.



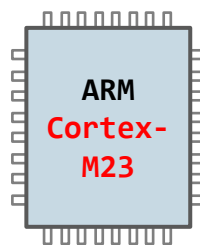
ARMv
6-M



ARMv
6-M



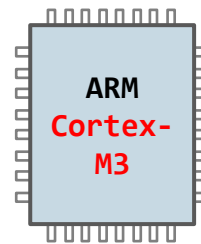
ARMv
6-M



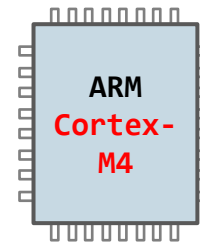
ARMv
8-M

Harvard

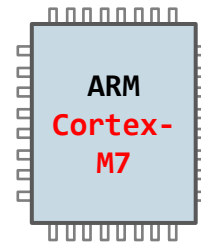
Data and instructions are stored into separate memories.



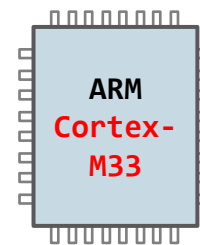
ARMv
7-M



ARMv7
E-M



ARMv7
E-M



ARMv
8-M

Levels of Program Code

C Program

```
int main(void){  
    int i;  
    int total = 0;  
    for (i = 0; i < 10;  
    i++) {  
        total += i;  
    }  
    while(1); // Dead loop  
}
```

? High-level language

- ? Level of abstraction closer to problem domain
- ? Provides for productivity and portability

Assembly Program

Compile

```
MOVSB r1, #0  
MOVSB r0, #0  
B    check  
loop ADD  r1, r1, r0  
    ADDS r0, r0, #1  
check CMP  r0, #10  
    BLT  loop  
self  B    self
```

? Assembly language

- ? Textual representation of instructions

Assemble

Machine Program

```
0010000100000000  
0010000000000000  
1110000000000001  
0100010000000001  
0001110001000000  
0010100000001010  
1101110011111011  
1011111100000000  
1110011111111110
```

? Hardware representation

- ? Binary digits (bits)
- ? Encoded instructions and data

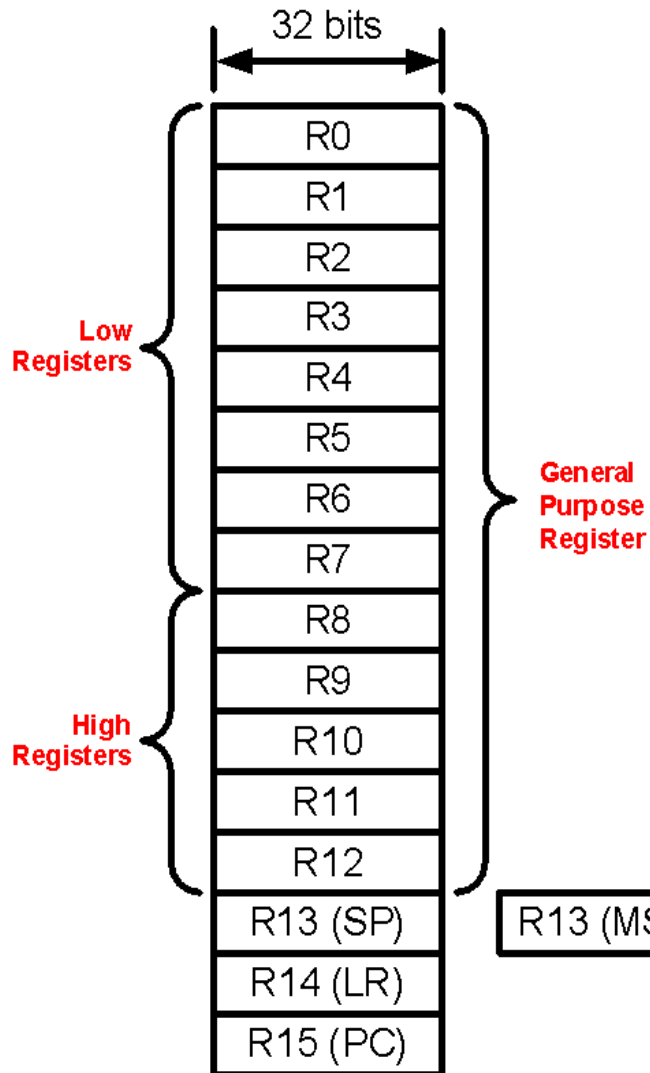
Memory

- linear addressing
- 2 descending stacks
 - main stack
 - process stack – used by OS
- all operations based on registers
- must move data between registers & memory

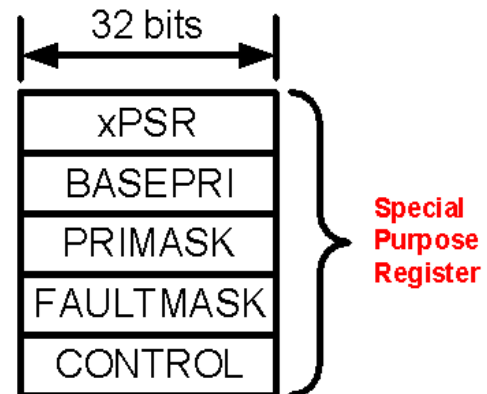
Registers

- 16 * 32 bit registers
- *working* registers
 - R0-R7 – *low* registers
 - R8-R12 – *high* registers
- R13 – stack pointer (SP)
- R14 - link register (LR)
- R15 – program counter (PC)
- PSR – program status register
 - bits for negative (N), zero (Z), carry (C), overflow (V)

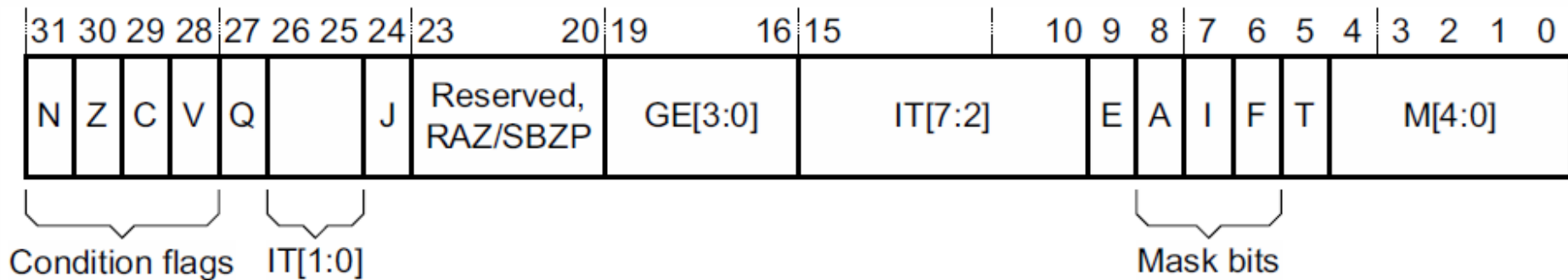
Registers-continued



- ? Fastest way to read and write
- ? Registers are within the processor chip
- ? A register stores 32-bit value
- ? STM32L has
 - ? **R0-R12**: 13 general-purpose registers
 - ? **R13**: Stack pointer (Shadow of MSP or PSP)
 - ? **R14**: Link register (LR)
 - ? **R15**: Program counter (PC)
 - ? Special registers (xPSR, BASEPRI, PRIMASK, etc)



Program Status Registers



❖ Condition code flags

- ❖ N = Negative result from ALU
- ❖ Z = Zero result from ALU
- ❖ C = ALU operation Carried out
- ❖ V = ALU operation oVerflowed

❖ Sticky Overflow flag - Q flag

- ❖ Architecture 5TE/J only
- ❖ Indicates if saturation has occurred

❖ J bit

- ❖ Architecture 5TEJ only
- ❖ J = 1: Processor in Jazelle state

❖ Interrupt Disable bits.

- ❖ I = 1: Disables the IRQ.
- ❖ F = 1: Disables the FIQ.

❖ T Bit

- ❖ Architecture xT only
- ❖ T = 0: Processor in ARM state
- ❖ T = 1: Processor in Thumb state

❖ Mode bits

- ❖ Specify the processor mode

Data formats

- *word*
 - 32 bits
 - long
- *half word*
 - 16 bits
 - int
- byte
 - 8 bits
 - char

Programming approach

- in high level programming:
 - identify variables & types
 - compiler maps these to memory & data representation
 - use of memory v registers invisible
- in low level programming
 - must make explicit use of memory & registers
 - must choose data representation
- registers much faster than memory
- try to map variables to registers
- what if more variables than registers...?

Data processing Instructions

- * **Largest family of ARM instructions, all sharing the same instruction format.**
- * **Contains:**
 - Arithmetic operations
 - Comparisons (no results - just set condition codes)
 - Logical operations
 - Data movement between registers
- * **Remember, this is a load / store architecture**
 - These instructions only work on registers, *NOT* memory.
- * **They each perform a specific operation on one or two operands.**
 - First operand always a register - Rn
 - Second operand sent to the ALU via barrel shifter.
- * **We will examine the barrel shifter shortly.**

Instruction format

instruction Rd, Rn, operand₂

- *instruction* is a mnemonic
 - meaningful short form
 - reminds us what instruction does
- *Rd* == destination register - R0..R15
- *Rn* == operand register - may be optional
- *operand₂* == *register* or *constant*
- *constant* == *#number*

Using and updating the Condition Field

- * **To execute an instruction conditionally, simply postfix it with the appropriate condition:**
 - For example an add instruction takes the form:
 - `ADD r0,r1,r2` ; `r0 = r1 + r2` (ADDAL)
 - To execute this only if the zero flag is set:
 - `ADDEQ r0,r1,r2` ; If zero flag set then...
; ... `r0 = r1 + r2`
- * **By default, data processing operations do not affect the condition flags (apart from the comparisons where this is the only effect). To cause the condition flags to be updated, the S bit of the instruction needs to be set by postfixing the instruction (and any condition code) with an “S”.**
 - For example to add two numbers and set the condition flags:
 - `ADDS r0,r1,r2` ; `r0 = r1 + r2`
; ... and set flags

Data Movement

* **Operations are:**

- MOV operand2
- MVN NOT operand2

Note that these make no use of operand1.

* **Syntax:**

- <Operation> {<cond>} {S} Rd, Operand2

* **Examples:**

- MOV r0, r1
- MOVS r2, #10
- MVNEQ r1, #0

MOV: move

MOV *Rd, operand2*

Rd = *operand2*

- don't update condition code flags

MOVS *Rd, operand2*

- as MOV but update condition code flags

MOV *Rd, #number*

Rd = *number*

NB *number* <= 255 - more later

MVN: move with logical NOT

`MVN Rd, operand2`

$Rd = \text{operand2}$

- don't update condition code flags
- performs a bitwise logical NOT operation on the operand2, and places the result into Rd.

`MVN Rd, #number`

$Rd = \neg \text{number}$

Label

- Any instruction can be associated with a label
- Example:
 - `start: ADD r0, r1, r2 ; a = b+c`
`next: SUB r1, r1, #1 ; b--`
- In fact, every instruction has a label regardless if the programmer explicitly names it
 - The label is the address of the instruction
 - A label is a pointer to the instruction in memory

Label

- Therefore, the text label doesn't exist in binary code
- relative to program counter (PC)
- turned into PC + offset in machine code
- may start with _ such as _global

System calls

- `SWI 0` == software interrupt to call function
- parameters in R0-R2
- system function number in R7
- e.g. `exit` is 1
- NB shell variable `$?` is R0 so:
`echo $?` == display R0

Layout

- usual to precede instruction with a *tab*
- comment is: *@ text // text*
- large comment: */*.....*/*

Program skeleton

```
.global _start
_start:
@ program code here
_exit:
    MOV R0, #65 @ arbitrary value
    MOV R7, #1
    SWI 0
```

- `.global _start`
 - assembler directive
 - entry point is visible externally
`_start`

Running assembler programs

- *file.s* == assembly language source

```
$ as -o file.o file.s
```

- assemble *file.s* to *file.o*

```
$ ld -o file file.o
```

- create executable *file* from *file.o*

```
$ ./file
```

- run executable in *file* from `_start`

```
$ echo $?
```

- display R0

Example: exit

- suppose `exit.s` is basic start program

```
$ as -o exit.o exit.s
```

```
$ ld -o exit exit.o
```

```
$ ./exit
```

```
$ echo $?
```

Arithmetic Operations

* Operations are:

- ADD $\text{operand1} + \text{operand2}$
- ADC $\text{operand1} + \text{operand2} + \text{carry}$
- SUB $\text{operand1} - \text{operand2}$
- SBC $\text{operand1} - \text{operand2} + \text{carry} - 1$
- RSB $\text{operand2} - \text{operand1}$
- RSC $\text{operand2} - \text{operand1} + \text{carry} - 1$

* Syntax:

- $\langle \text{Operation} \rangle \{ \langle \text{cond} \rangle \} \{ S \} \text{ Rd, Rn, Operand2}$

* Examples

- ADD r0, r1, r2
- SUBGT r3, r3, #1
- RSBLES r4, r5, #5

ADD: addition

ADD *Rd, Rn, operand₂*

Rd = *Rn* + *operand₂*

ADD *Rd, operand₂*

Rd = *Rd* + *operand₂*

ADDC

- add with carry
- like ADD + carry flag

ADDS/ADDCS

- like ADD/ADDC but set condition code flags

SUB: subtraction

SUB Rd, Rn, operand₂

Rd = Rn - operand₂

SUB Rd, operand₂

Rd = Rd - operand₂

SBC

- subtract with carry
- like *SUB -1* if carry flag not set

SUBS/SBCS

- like *SUB/SUBC* but set condition code flags

RSB: reverse subtraction

RSB *Rd, Rn, operand₂*

Rd = *operand₂* - ***Rn***

RSB *Rd, operand₂*

Rd = *operand₂* - *Rd*

Multiplication

- In ARM, we multiply registers, so:
 - 32-bit value x 32-bit value = 64-bit value
- Syntax of Multiplication (signed):
 - MUL register1, register2, register3
- Built in multiplication MUL
 - MUL *Rd*, *Rn*, *operand*₂
 - *Rd* = *Rn***operand*₂
 - NB *Rd* cannot be an *operand*

Multiplication

❖ Example:

❖ in C: `a = b * c;`

❖ in MIPS:

❖ let b be r2; let c be r3; and let a be r0 and r1 (since it may be up to 64 bits)

`MUL r0, r2, r3 ; b*c only 32 bits
stored`

Note: Often, we only care about the lower half of the product.

`SMULL r0, r1, r2, r3 ; 64 bits in r0:r1`

Multiply and Divide

- ❖ There are 2 classes of multiply - producing 32-bit and 64-bit results
- ❖ 32-bit versions on an ARM7TDMI will execute in 2 - 5 cycles

- ❖ `MUL r0, r1, r2` ; `r0 = r1 * r2`

- ❖ `MLA r0, r1, r2, r3` ; `r0 = (r1 * r2) + r3`

- ❖ 64-bit multiply instructions offer both signed and unsigned versions
 - ❖ For these instruction there are 2 destination registers

- ❖ `[U|S]MULL r4, r5, r2, r3` ; `r5:r4 = r2 * r3`

- ❖ `[U|S]MLAL r4, r5, r2, r3` ; `r5:r4 = (r2 * r3) + r5:r4`

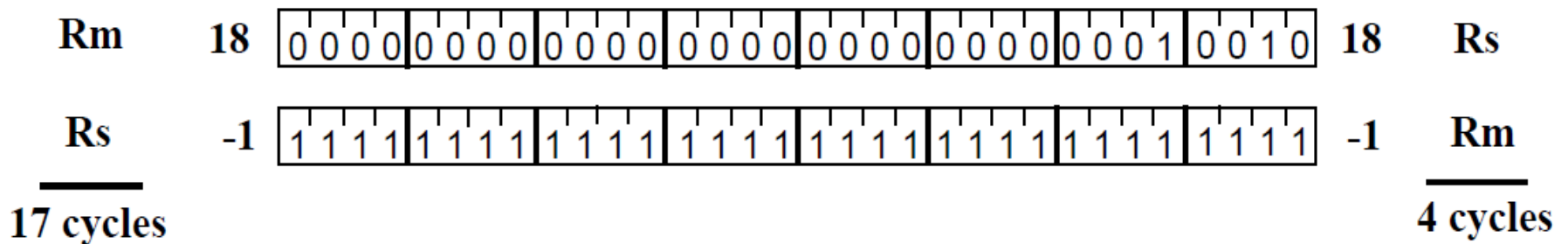
- ❖ Most ARM cores do not offer integer divide instructions
 - ❖ Division operations will be performed by C library routines or inline shifts

Multiplication Instructions

- * **The Basic ARM provides two multiplication instructions.**
 - * **Multiply**
 - $MUL\{<cond>\}\{S\} Rd, Rm, Rs$; $Rd = Rm * Rs$
 - * **Multiply Accumulate - does addition for free**
 - $MLA\{<cond>\}\{S\} Rd, Rm, Rs, Rn$; $Rd = (Rm * Rs) + Rn$
 - * **Restrictions on use:**
 - Rd and Rm cannot be the same register
 - Can be avoid by swapping Rm and Rs around. This works because multiplication is commutative.
 - Cannot use PC.
- These will be picked up by the assembler if overlooked.**
- * **Operands can be considered signed or unsigned**
 - Up to user to interpret correctly.

Multiplication Implementation

- * **The ARM makes use of Booth's Algorithm to perform integer multiplication.**
- * **On non-M ARMs this operates on 2 bits of Rs at a time.**
 - For each pair of bits this takes 1 cycle (plus 1 cycle to start with).
 - However when there are no more 1's left in Rs, the multiplication will early-terminate.
- * **Example: Multiply 18 and -1 : $Rd = Rm * Rs$**



- * Note: Compiler does not use early termination criteria to decide on which order to place operands.**

Extended Multiply Instructions

- * **M variants of ARM cores contain extended multiplication hardware. This provides three enhancements:**
 - *An 8 bit Booth's Algorithm is used*
 - Multiplication is carried out faster (maximum for standard instructions is now 5 cycles).
 - *Early termination method improved so that now completes multiplication when all remaining bit sets contain*
 - all zeroes (as with non-M ARMs), or
 - all ones.

Thus the previous example would early terminate in 2 cycles in both cases.

- *64 bit results* can now be produced from two 32bit operands
 - Higher accuracy.
 - Pair of registers used to store result.

Multiply-Long and Multiply-Accumulate Long

- * **Instructions are**
 - MULL which gives $RdHi, RdLo := Rm * Rs$
 - MLAL which gives $RdHi, RdLo := (Rm * Rs) + RdHi, RdLo$
- * **However the full 64 bit of the result now matter (lower precision multiply instructions simply throws top 32bits away)**
 - Need to specify whether operands are signed or unsigned
- * **Therefore syntax of new instructions are:**
 - UMULL {<cond>} {S} RdLo, RdHi, Rm, Rs
 - UMLAL {<cond>} {S} RdLo, RdHi, Rm, Rs
 - SMULL {<cond>} {S} RdLo, RdHi, Rm, Rs
 - SMLAL {<cond>} {S} RdLo, RdHi, Rm, Rs
- * **Not generated by the compiler.**

Warning : Unpredictable on non-M ARMs.

Instruction Summary So far

❖ In ARM Assembly Language:

- ❖ Registers replace C variables
- ❖ One Instruction (simple operation) per line
- ❖ Simpler is Better
- ❖ Smaller is Faster

❖ Instructions so far:

ADD, SUB, RSB, MUL, MULA, [U|S]MULL,
[U|S]MLAL

❖ Registers:

Places for general variables: r0–r12

A Sample program

```
@.org 0x1000 // Start at memory location 1000

.data

numbers:

    .word 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30

num_count:

    .word 30

.text // Code section

.global _start

_start:

@ Initialize the sum to 0

    mov r0, #0

@ Load the number of elements (30) into r1

    ldr r1, =num_count

    ldr r1, [r1]

@ Load the address of the numbers array into r2

    ldr r2, =numbers

|
```

A Sample program (continue)

loop:

@ Load the current number into r3

ldr r3, [r2], #4 @ Load the value at the current address and increment the address by 4 bytes

@ Add the current number to the sum in r0

add r0, r0, r3

@ Decrement the counter (r1) by 1

subs r1, r1, #1

@ Check if the counter has reached 0 (end of loop)

bne loop @ Branch to 'loop' if r1 is not equal to 0

@ The loop is done; r0 contains the sum

@ Put your further code here or return from the function if needed

@ For example, to exit a program in ARM assembly, you can use the following:

mov r7, #0 @ syscall number for "exit"

// End program

_stop:

B _stop

.end

Expressions

- for:

$var_1 = var_2 \text{ op } var_3$

- with:

$var_1 \text{ in } R_1$

$var_2 \text{ in } R_2$

$var_3 \text{ in } R_3$

$\text{op } R_1, R_2, R_3$

- e.g.

$x = y * z ;$

@ x == R1

@ y == R2

@ z == R3

MUL R1, R2, R3

Expressions

- for:

$var_1 = var_2 op_1 var_3 op_2 var_4$

- if op_1 and op_2 have same precedence
- or op_1 has higher precedence than op_2 ☐

$op_1 R_1, R_2, R_3$

$op_2 R_1, R_4$

e.g.

$x = y * z - a;$

...

@ a == R4

MUL R1, R2, R3

SUB R1, R4

Expressions

- for:

$var_1 = var_2 op_1 var_3 op_2 var_4$

- if op_2 has higher precedence than op_1

- must evaluate op_2 expression in new register[?]

$op_2 R_i, R_3, R_4$

$op_1 R_1, R_2, R_i$

e.g.

$x = y + z * a;$

MUL R5, R3, R4

ADD R1, R2, R5

e.g.

$x = y * (z + a)$

ADD R5, R3, R4

MUL R1, R2, R5

Comparisons

- * **The only effect of the comparisons is to**
 - UPDATE THE CONDITION FLAGS. Thus no need to set S bit.
- * **Operations are:**
 - CMP operand1 - operand2, but result not written
 - CMN operand1 + operand2, but result not written
 - TST operand1 AND operand2, but result not written
 - TEQ operand1 EOR operand2, but result not written
- * **Syntax:**
 - <Operation>{<cond>} Rn, Operand2
- * **Examples:**
 - CMP r0, r1
 - TSTEQ r2, #5

CMP: compare

CMP *Rd*, *operand*₂

- subtract *operand*₂ from *Rd* BUT ...
- ... do not modify *Rd*
 - otherwise same as SUBS
- update Z, N, C, V flags

CMN *Rd*, *operand*₂

- add *operand*₂ to *Rd* BUT ...
- ... do not modify *Rd*
 - otherwise same as ADDS
- update Z, N, C, V flags

Flags

- N – 1 if result <0; 0 otherwise
- Z – 1 if result =0; 0 otherwise
- C – 1 if result led to carry; 0 otherwise
 - i.e. $X+Y > 2^{32}$
 - i.e. $X-Y \geq 0$

Flags

- V – 1 if result led to overflow; 0 otherwise
 - (-) == negative
 - (+) == positive
 - i.e. $(-)X + (-)Y > 0$
 - i.e. $(+)X + (+)Y < 0$
 - i.e. $(-)X - (+)Y > 0$
 - i.e. $(+X) - (-Y) < 0$

B: branch

B *label*

- branch to *label*
- i.e. reset PC to address for *label*

B *cond label*

- branch on *condition* to *label*

C Decisions: `if` Statements

- ❖ `if` statements in C

- ❖ `if (condition) clause`

- ❖ `if (condition) clause1 else clause2`

- ❖ Rearrange 2nd `if` into following:

- `if (condition) goto L1;`

- `else goto`

- `L2; L1: clause1;`

- `L2: clause2;`

- ❖ Not as elegant as `if-else`, but same meaning

ARM goto Instruction Conversion

- ❖ The simplest control instruction is equivalent to a C `goto` statement
- ❖ `goto label` (in C) is the same as:
- ❖ `B label` (in ARM)
- ❖ `B` is shorthand for “branch”. This is called an unconditional branch meaning that the branch is done regardless of any conditions.
- ❖ There are also conditional branches

Condition

suffix	flags
EQ == equal	Z=1
NE == not equal	Z=0
CS/HS == carry set/ higher or same - unsigned	C=1
CC/LO == carry clear/lower - unsigned	C=0
MI == negative	N=1
PL == positive or 0	N=0
VS == overflow	V=1
VC == no overflow	V=0

Condition

suffix	flags
HI == higher - unsigned	C=1 & Z=0
LS == lower or same - unsigned	C=0 or Z=1
GE == greater than or equal - signed	N=V
LT == less than - signed	N!=V
GT == greater than - signed	Z=0 & N=V
LE == less than or equal, signed	Z=1 or N!=V
AL == any value	default if not <i>cond</i>

Overflow Detection

One way to detect overflow is to check whether the sign bit is consistent with the sign of the inputs when the two inputs are of the same sign – if you added two positive numbers and got a negative number, something is wrong, and vice versa.

A+B			
Sign of A	Sign of B	Sign of Result	Overflow?
+	+	-	Yes
-	-	+	
+	+	+	No
-	-	-	
+	-	-	No
+	-	+	No
-	+	-	No
-	+	+	No

ARM BRANCH Instructions

- ❖ ARM also has variants of the branch instruction that only goto the label if a certain condition is TRUE
- ❖ Examples:
 - ❖ `BEQ label ; BRANCH EQUAL`
 - ❖ `BNE label ; BRANCH NOT EQUAL`
 - ❖ `BLE label ; BRANCH LESS THAN EQUAL`
 - ❖ `BLT label ; BRANCH LESS THAN`
 - ❖ `BGE label ; BRANCH GREATER THAN EQUAL`
 - ❖ `BGT label ; BRANCH GREATER THAN`
 - ❖ Plus more ...
- ❖ The condition is T/F based upon the fields in the Program Status Register

Example: multiply by adding

```
int x;  
int y;  
int m;  
x = 3;  
y = 10;  
m = 0;  
while(y!=0)  
{  
    m = m+x;  
y = y-1;  
}
```

```
.global _start  
_start:  
    MOV R1, #0x03  
    MOV R2, #0x0a  
    MOV R3, #0x00  
_loop:  
    CMP R2, #0x00  
    BEQ _exit  
    ADD R3, R1  
    SUB R2, #0x01  
    B _loop
```

Example: multiply by adding

```
#m=x*y
int x;
int y;
int m;

x = 8;
y = 10;
m = 0;
while(y!=0)
{
    m = m+x;
    y = y-1;
}
```

```
_exit:
    MOV R0, R3
    MOV R7, #1
    SWI 0

...
$ echo $?
1E
```

Data directives & constants

`.data`

- start of sequence of data directives
- usually after instructions program

`.equ label, value`

- define constant
- associate *label* with *value*
- use `#label` as *operand*₂ to get *value*

Example: multiply by adding

```
•m == x*y
int x;
int y;
int m;
x = 8;
y = 10;
m = 0;
while(y!=0)
{
    m =
    m+x;    y
    = y-1;
}
```

```
.global _start
_start:
    MOV R1, #X
    MOV R2, #Y
    MOV R3, #0x00
_loop:
    CMP R2, #0x00
    BEQ _exit
    ADD R3, R1
    SUB R2, #0x01
    B _loop
```

Example: multiply by adding

```
m=x*y
    int x;
    int y;
    int m;
    x = 8;
    y = 10;
    m = 0;
    while(y!=0)
    {   m = m+x;
        y = y-1;
    }
```

```
_exit:
    MOV R0, R3
    MOV R7, #1
    SWI 0

.data
.equ X, 3
.equ Y, 4
...
$ echo $?
12
```

Tracing and debugging

- `gdb` - GNU debugger
- assemble with `-g` flag
- run program from `gdb`

```
$ as -g -o file.o file.s
```

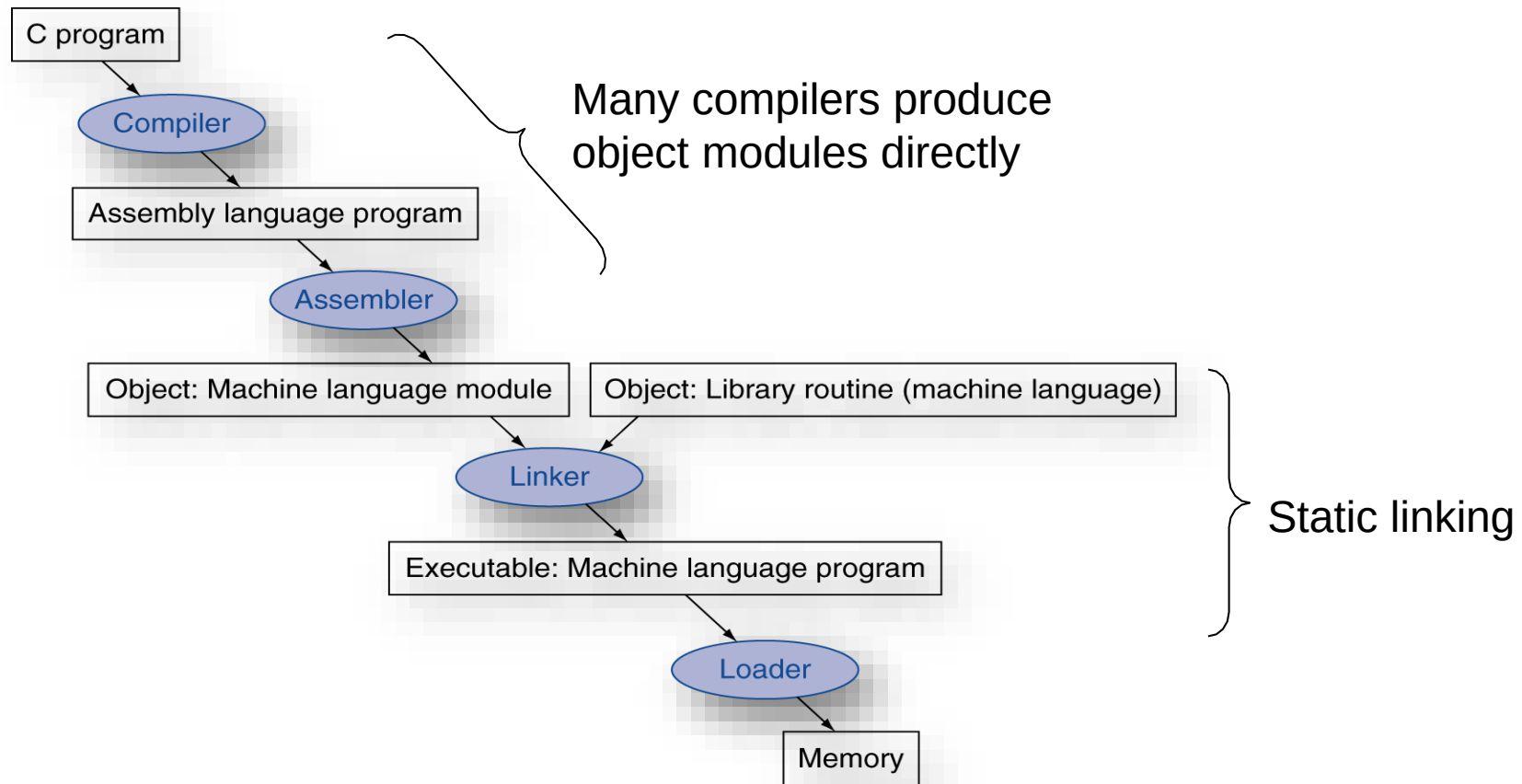
```
$ ld -o file file.o
```

```
gdb file
```

```
...
```

```
(gdb) command
```

Translation and Startup



UNIX: C source files are named x.c, assembly files are x.s, object files are named x.o, statically linked library routines are x.a, dynamically linked library routes are x.so, and executable files by default are called a.out.

MS-DOS uses the .C, .ASM, .OBJ, .LIB, .DLL, and .EXE to the same effect.

Tracing and debugging

- `q(uit)` - return to shell
- `l(ist)` - show program - may need to press return
- `r(un)` - run program
- `b(reak) number` – set breakpoint at line *number*
 - program pauses at breakpoint
 - can set multiple breakpoints
 - can also set break points at labels & addresses

Tracing and debugging

- `i(nfo) r` – show registers
- `c(continue)` - continue execution after breakpoint
- `s tep)` - execute next instruction
- `i(info) b` - show breakpoints
- `d(delete) number` – remove breakpoint *number*

Tracing and debugging

- to trace
 - set breakpoint at start of program
 - step & show registers
- to debug
 - set breakpoints at salient points in program
 - e.g. at start or ends of loops
 - continue/step & show registers

Tracing and debugging

- e.g. suppose mult program is in

mult.s

```
$ as -g -o mult.o mult.s
```

```
$.ld -o mult mult.s
```

```
$ gdb mult
```

```
(gdb)
```

```
1      .global _start
```

```
2
```

```
3 _start:
```

```
4      MOV R1, #X
```

```
5      MOV R2, #Y
```

```
6      MOV R3, #0x00
```

```
7 _loop:
```

```
8      CMP R2, #0x00
```

```
9      BEQ _exit
```

```
10     ADD R3, R1
```

```
(gdb)
```

```
11     SUB R2, #0x01
```

```
12     B    _loop
```

```
13 _exit:
```

```
14     MOV R0, R3
```

```
15     MOV R7, #1
```

```
16     SWI 0
```

```
17
```

```
18 .data
```

```
19 .equ X, 3
```

```
20 .equ Y, 4
```

```
(gdb)
```

Tracing & debugging

```
(gdb) b _start
Breakpoint 1 at 0x805b:
file mult.s, line 5
(gdb) r
...
5      MOV R2, #Y
(gdb) i r
r0      0x0      0
r1      0x3      3
r2      0x0      0
r3      0x0      0
(gdb) s
6      MOV R, #0
```

```
(gdb) i r
r0      0x0      0
r1      0x3      3
r2      0x0      4
r3      0x0      0
...
(gdb) b 12
Breakpoint 2 at 0x8070:
file mult.s, line 12
(gdb) c
...
12      B _loop
```

Tracing & debugging

```
(gdb) i r
r0      0x0      0
r1      0x3      3
r2      0x0      3
r3      0x0      3
...
(gdb) c
...
12      B _loop
(gdb) i r
r0      0x0      0
r1      0x3      3
r2      0x0      2
r3      0x0      6
...
```

```
(gdb) c
...
12      B _loop
(gdb) i r
r0      0x0      0
r1      0x3      3
r2      0x0      1
r3      0x0      9
...
(gdb) c
...
12      B _ loop
```

Tracing & debugging

```
(gdb) i r
r0      0x0      0
r1      0x3      3
r2      0x0      0
r3      0x0      12
...
(gdb) c
...
[Inferior 1 (process 2614)
exited with code 014]
(gdb) q
$
```

NOP: no operation

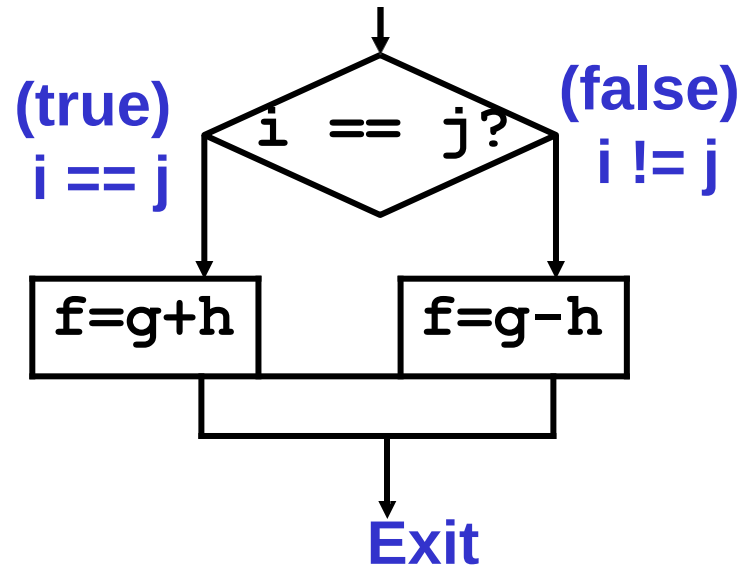
NOP

- do nothing
- use for padding/layout
- no cost

Compiling C `if` into ARM

❖ Compile by hand

```
if (i == j) f=g+h;  
else  
f=g-h;
```



❖ Use this mapping:

`f`: r0, `g`: r1, `h`: r2, `i`: r3, `j`: r4

Comparison Instructions

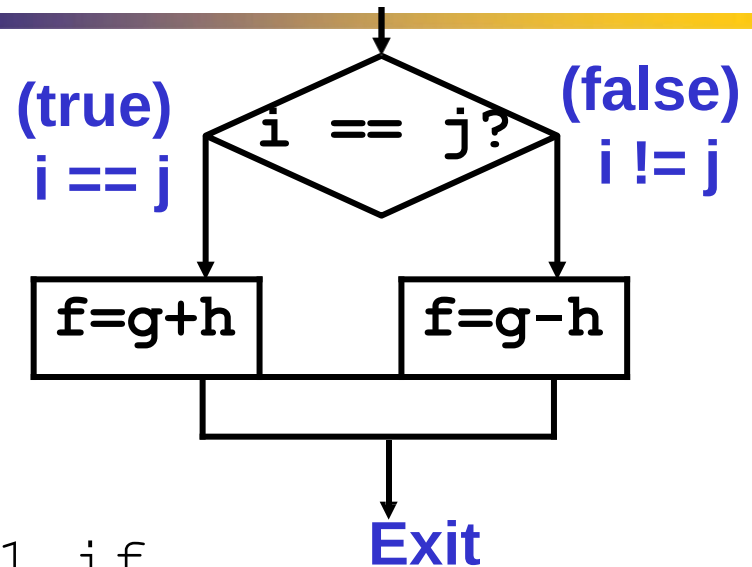
- ❖ In order to perform branch on the “==” operation we need a new instruction
- ❖ CMP – Compare: subtracts a register or an immediate value from a register value and updates condition codes
- ❖ Examples:
 - ❖ `CMP r3, #0 ; set Z flag if r3 == 0`
 - ❖ `CMP r3, r4 ; set Z flag if r3 == r4`

All flags are set as result of this operation, not just Z.

Compiling C `if` into ARM

Compile by hand

```
if (i == j) f=g+h;  
else f=g-h;
```



❖ Final compiled ARM code:

```
CMP    r3, r4        ; Z = 1 if  
i==j   BEQ True      ; goto True when i==j  
SUB    r0, r1, r2     ; f=g-h (false)  
B      Fin           ; goto Fin  
True:  ADD    r0, r1, r2 ; f=g+h (true)
```

Fin

Note: Compiler automatically creates labels to handle decisions (branches) appropriately. Generally not found in C code.

Loops in C/Assembly

❖ Simple loop in C;

```
do{  
    g--;  
    i = i + j;}  
while (i != h);
```

❖ Rewrite this as:

```
Loop: g--;  
i = i + j;  
if (i != h) goto Loop;
```

❖ Use this mapping:

```
g: r1, h: r2, i: r3, j:r4
```

Loops in C/Assembly

❖ Final compiled code:

```
Loop    SUB    r1, r1, #1      ; g--
        ADD    r3, r3, r4      ; i=i+j
        CMP    r3, r2          ; cmp i, h
        BNE    Loop           ; goto Loop
                                   ; if i!=h
```

Inequalities in ARM

- ❖ Until now, we've only tested equalities (`==` and `!=` in C). General programs need to test `<` and `>` as well.
- ❖ Use `CMP` and `BLE`, `BLT`, `BGE`, `BGT`
- ❖ Examples:

```
if (f < 10) goto Loop; =>  CMP r0, #10
                           BLT Loop
if (f >= i) goto Loop; =>  CMP r0, r3
                           BGE Loop
```

Loops in C/Assembly

- ❖ There are three types of loops in C:
 - ❖ `while`
 - ❖ `do... while`
 - ❖ `for`
- ❖ Each can be rewritten as either of the other two, so the method used in the previous example can be applied to `while` and `for` loops as well.
- ❖ **Key Concept:** Though there are multiple ways of writing a loop in ARM, conditional branch is _____key to decision making

Example: The C Switch Statement

- ❖ Choose among four alternatives depending on whether `k` has the value 0, 1, 2 or 3.

Compile this C code:

```
switch (k) {  
    case 0: f=i+j; break; /* k=0 */  
    case 1: f=g+h; break; /* k=1 */  
    case 2: f=g-h; break; /* k=2 */  
    case 3: f=i-j; break; /* k=3 */  
}
```

Example: The C Switch Statement

- ❖ This is complicated, so **simplify**.
- ❖ Rewrite it as a chain of if-else statements, which we already know how to compile:

```
if (k==0)  f=i+j;  
    else if (k==1)  f=g+h;  
        else if (k==2)  f=g-h;  
            else if (k==3)  f=i-j;
```

- ❖ Use this mapping:

```
f:  $r0,  g:  $r1,  h:  $r2,  i:  $r3,  
j:  r4,  k:  $r5
```

Example: The C Switch Statement

```
CMP  r5, #0           ; compare k, 0
BNE  L1               ; branch k!=0
ADD  r0, r3, r4       ; k==0 so f=i+j
B    Exit             ; end of case so Exit
L1   CMP  r5, #1       ; compare k, 1
BNE  L2               ; branch k!=1
ADD  r0, r1, r2       ; k==1 so f=g+h
B    Exit             ; end of case so Exit
L2   CMP  r5, #2       ; compare k, 2
BNE  L3               ; branch k!=2
SUB  r0, r1, r2       ; k==2 so f=g-h
B    Exit             ; end of case so Exit
L3   CMP  r5, #3       ; compare k, 3
BNE  Exit             ; branch k!=3
      SUB  r0, r3, r4   ; k==3 so f=i-j
Exit:
```

Predicated Instructions

- ❖ All instructions can be executed conditionally.
Simply add {EQ,NE,LT,LE,GT,GE, etc.} to end

C source code

```
if (r0 == 0)
{
    r1 = r1 + 1;
}
else
{
    r2 = r2 + 1;
}
```

ARM instructions

unconditional

```
CMP r0, #0
BNE else
ADD r1, r1, #1
B end
else:
    ADD r2, r2, #1
end
...
```

- 5 instructions
- 5 words
- 5 or 6 cycles

conditional

```
CMP r0, #0
ADDEQ r1, r1, #1
ADDNE r2, r2, #1
...
```

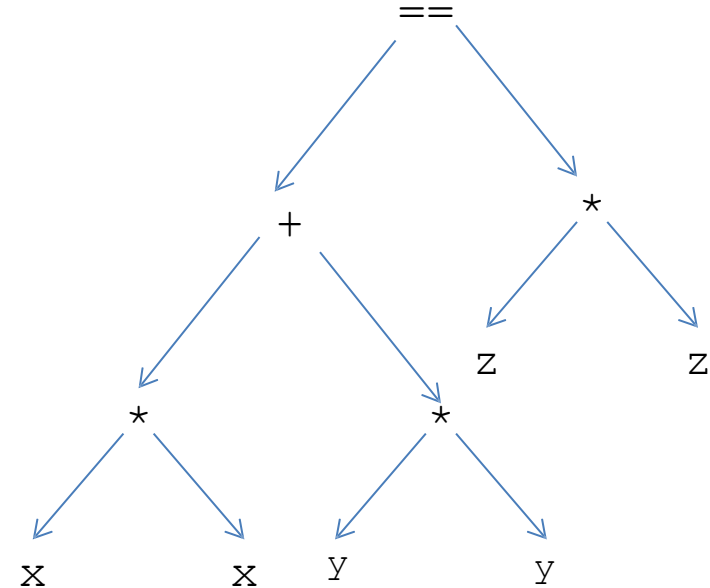
- 3 instructions
- 3 words
- 3 cycles

Expressions

- draw expression tree
- allocate registers to nodes
 - from bottom left
 - if expression in assignment then start with register for destination variable
- accumulate into register for left operand
- can re-use any register whose value is no longer required

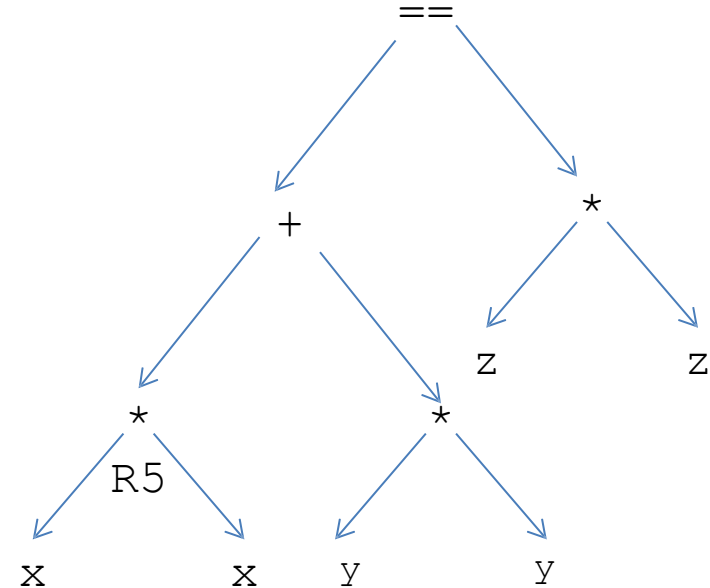
Example: Pythagoras

```
int x; /* R1 */
int y; /* R2 */
int z; /* R3 */
int p; /* R4 */
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```



Example: Pythagoras

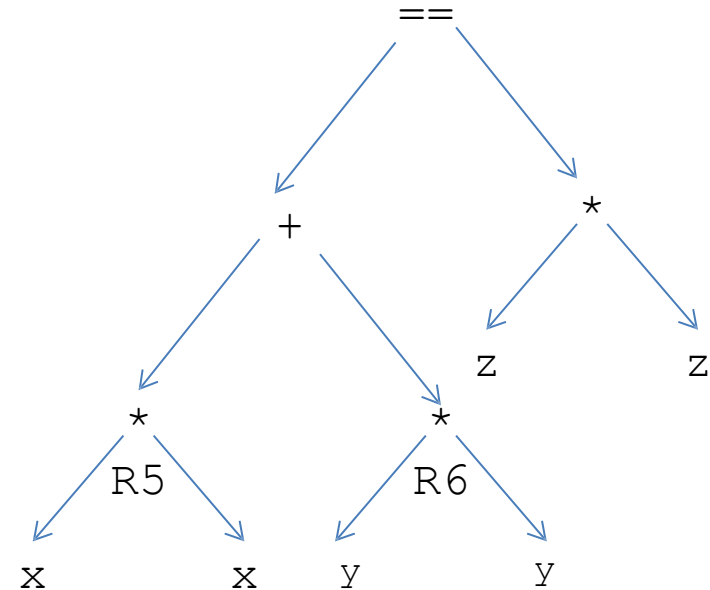
```
int x; /* R1 */
int y; /* R2 */
int z; /* R3 */
int p; /* R4 */
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```



Example:

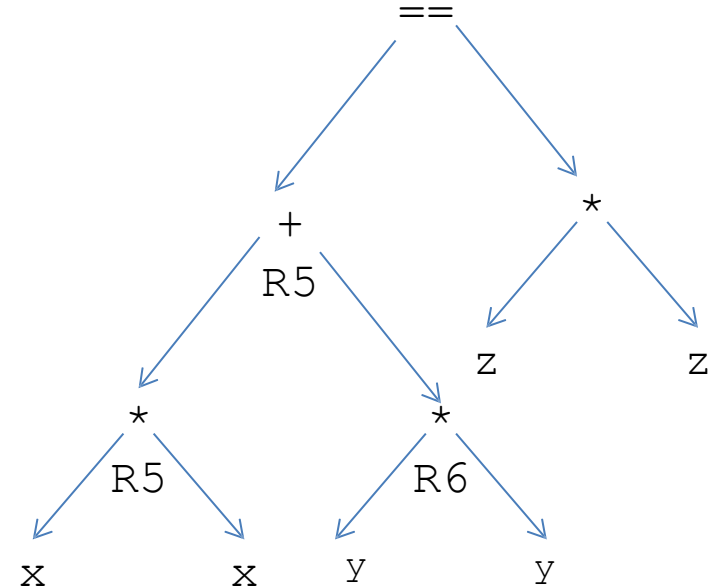
Pythagoras

```
int x; /* R1 */
int y; /* R2 */
int z; /* R3 */
int p; /* R4 */
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```



Example: Pythagoras

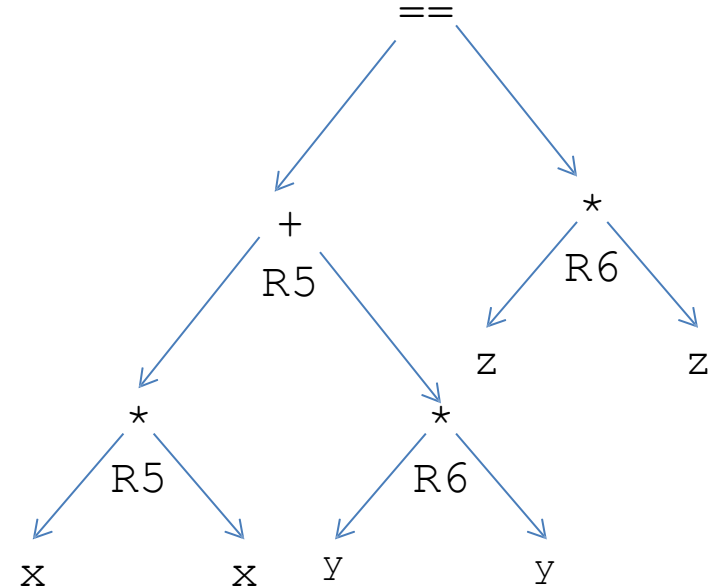
```
int x; /* R1 */
int y; /* R2 */
int z; /* R3 */
int p; /* R4 */
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```



Example:

Pythagoras

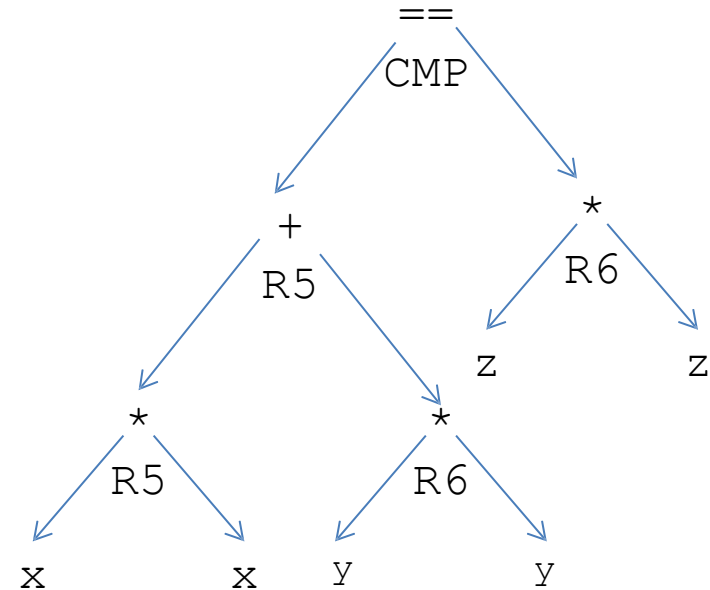
```
int x; /* R1 */
int y; /* R2 */
int z; /* R3 */
int p; /* R4 */
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```



Example:

Pythagoras

```
int x; /* R1 */
int y; /* R2 */
int z; /* R3 */
int p; /* R4 */
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```



Example:

Pythagoras

```
int x;
int y;
int z;
int p;
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;

.global _start
_start:
    MOV R1, #X
    MOV R2, #Y
    MOV R3, #Z
    MUL R5, R1, R1
    MUL R6, R2, R2
    ADD R5, R6
    MUL R6, R3, R3
    CMP R5, R6
    BEQ _same
    MOV R4, #0
    B _exit

_same:
    MOV R4, #1
```

Example:

Pythagoras

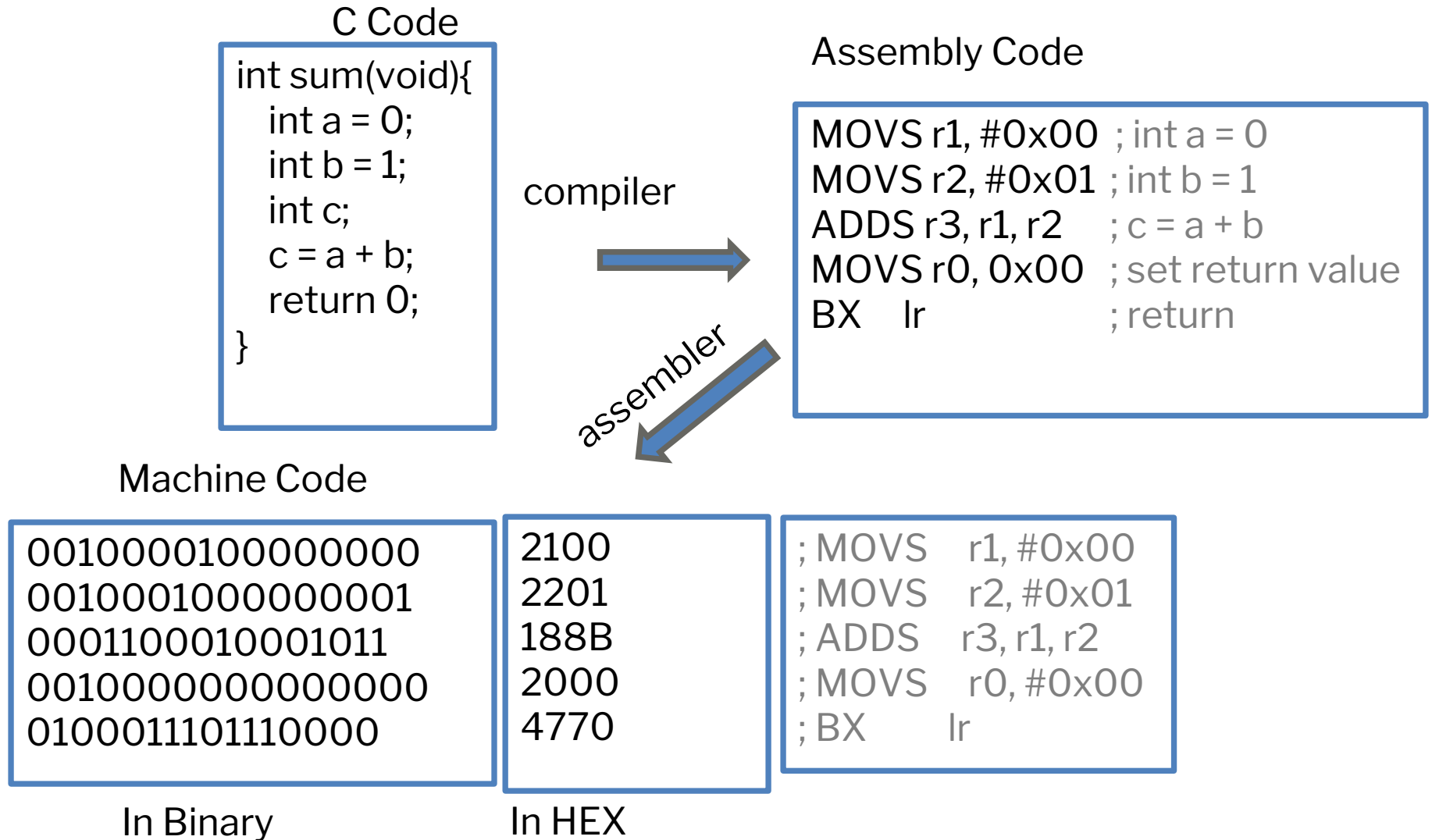
```
int x;
int y;
int z;
int p;
x = 3;
y = 4;
z = 5;
if (x*x+y*y==z*z)
    p=1;
else
    p=0;
```

```

_exit:
MOV R0, R4
MOV R7, #1
SWI 0

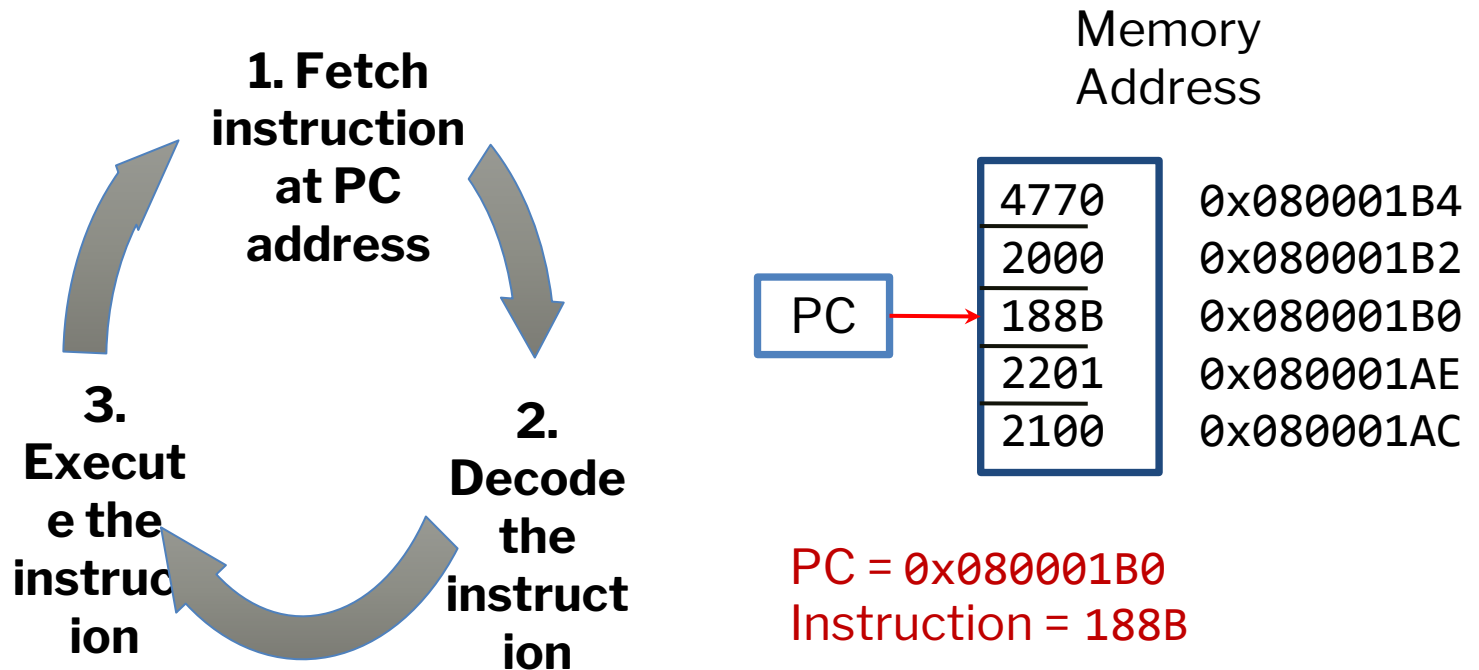
.data
.equ X, 3
.equ Y, 4
.equ Z, 5
```

See how a Program Runs



Program Execution

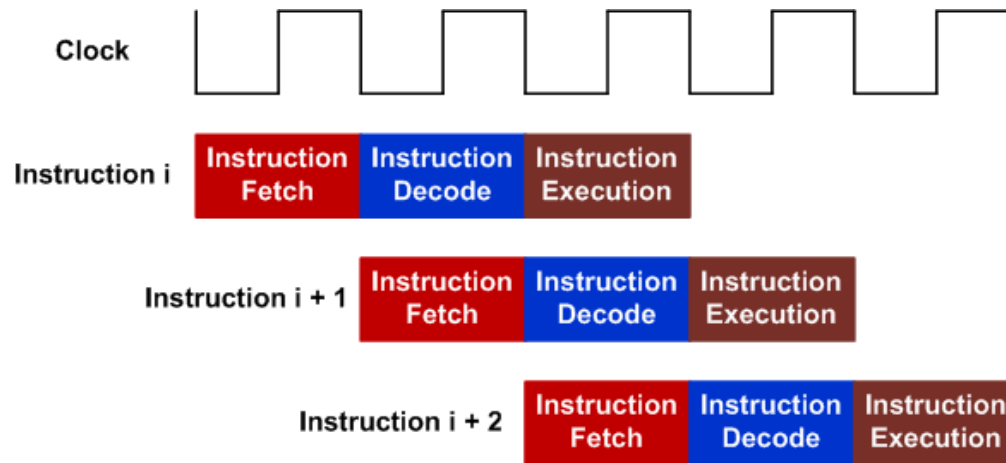
Program Counter (PC) is a register that holds the memory address of the next instruction to be fetched from the memory.



Three-state pipeline: Fetch, Decode, Execution

Pipelining allows hardware resources to be fully utilized

One 32-bit instruction or two 16-bit instructions can be fetched.

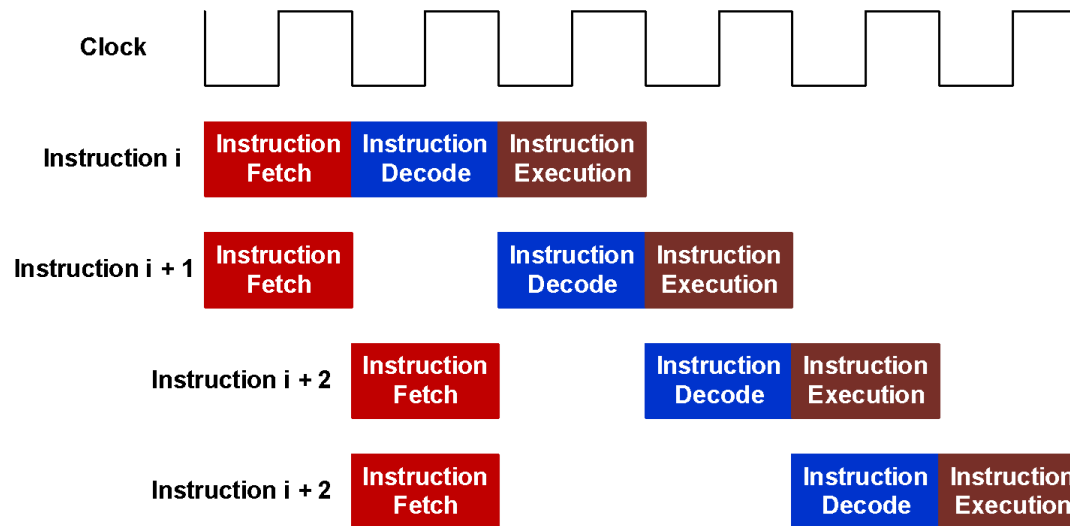


Pipeline of 32-bit instructions

Three-state pipeline: Fetch, Decode, Execution

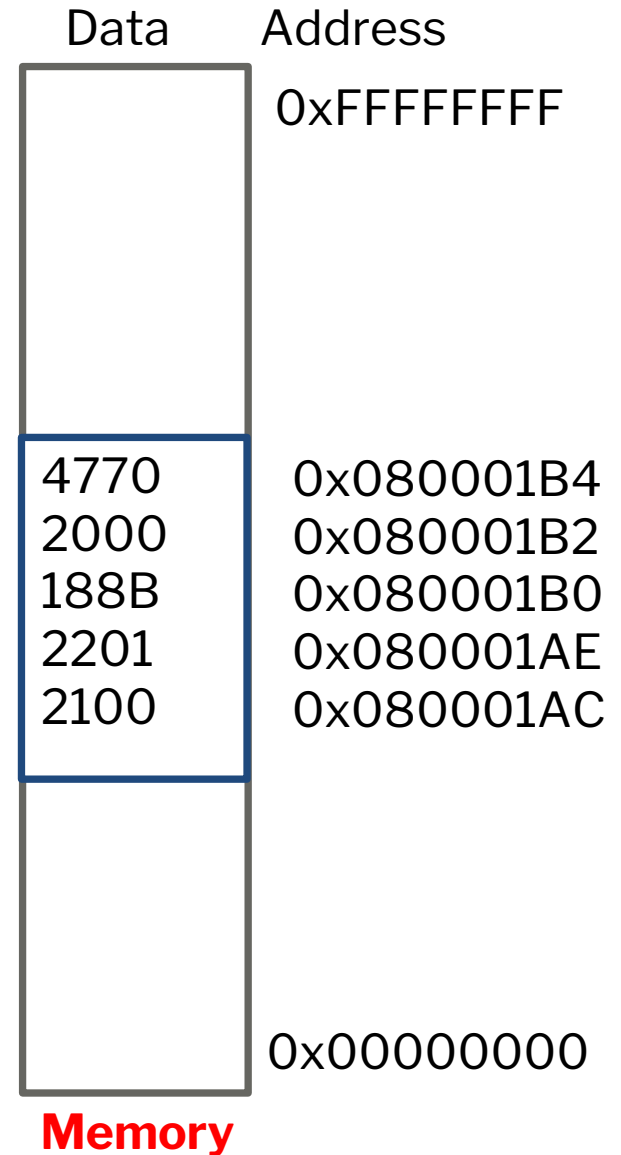
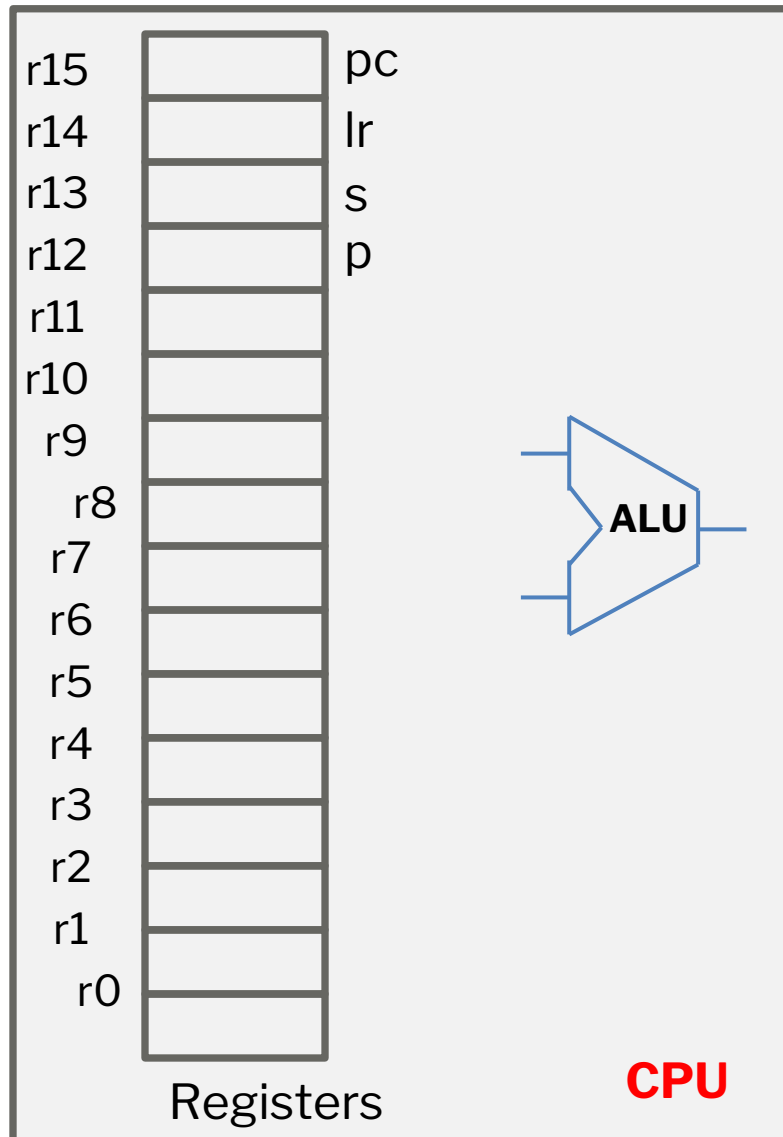
Pipelining allows hardware resources to be fully utilized

One 32-bit instruction or two 16-bit instructions can be fetched.



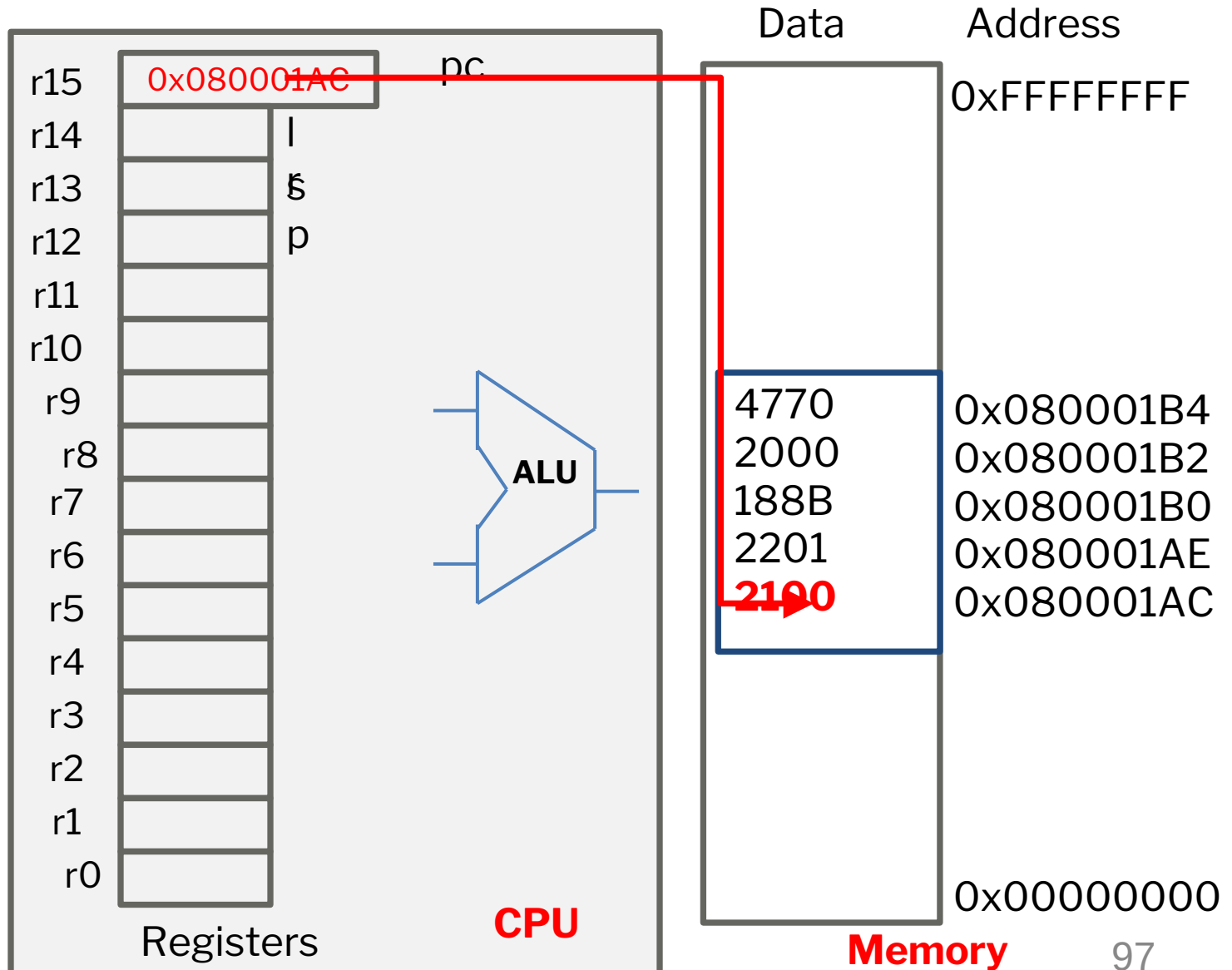
Pipeline of 16-bit instructions

Machine codes are stored in memory

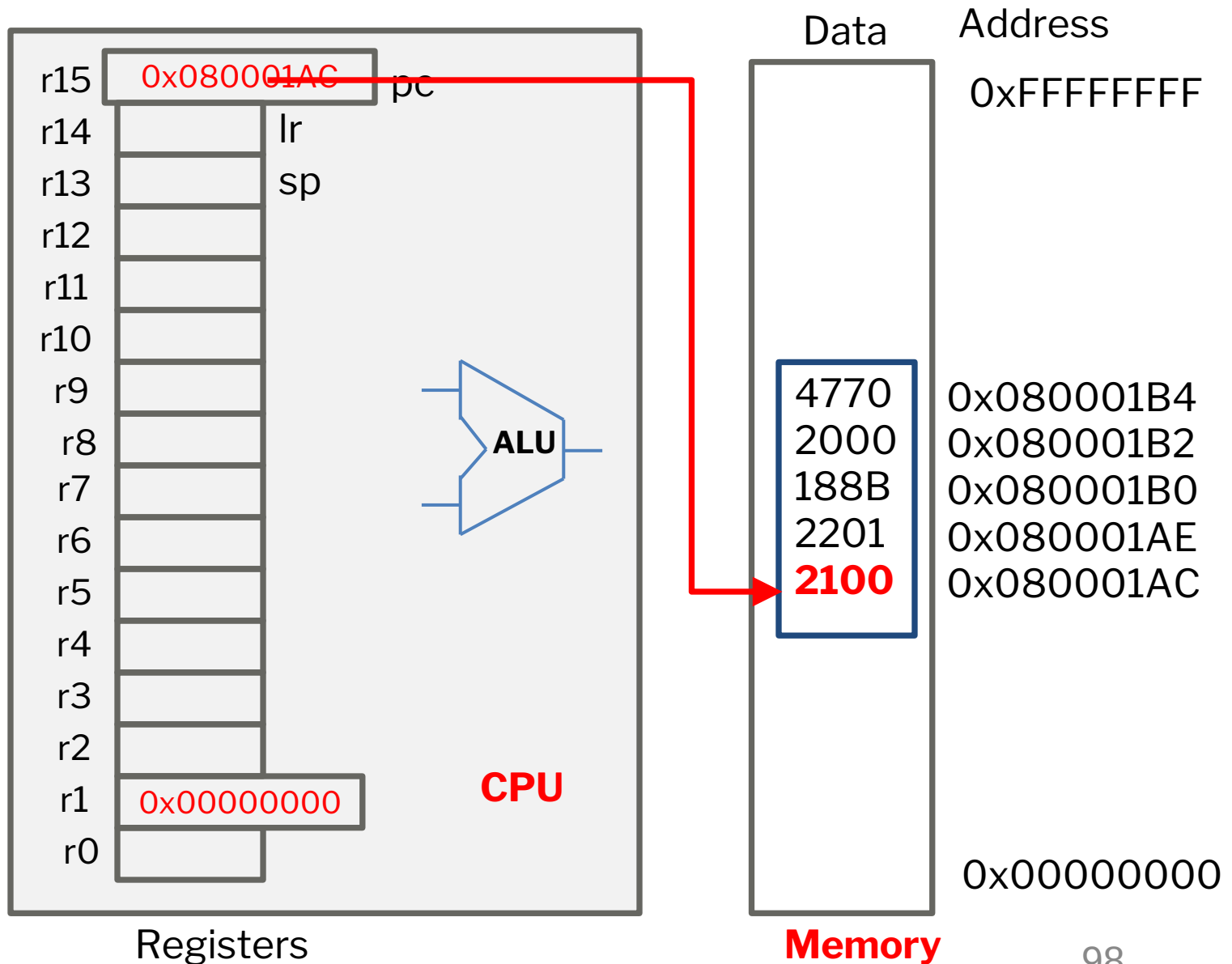


Fetch Instruction: pc = 0x08001AC

Decode Instruction: 2100 = **MOVS r1, #0x00**

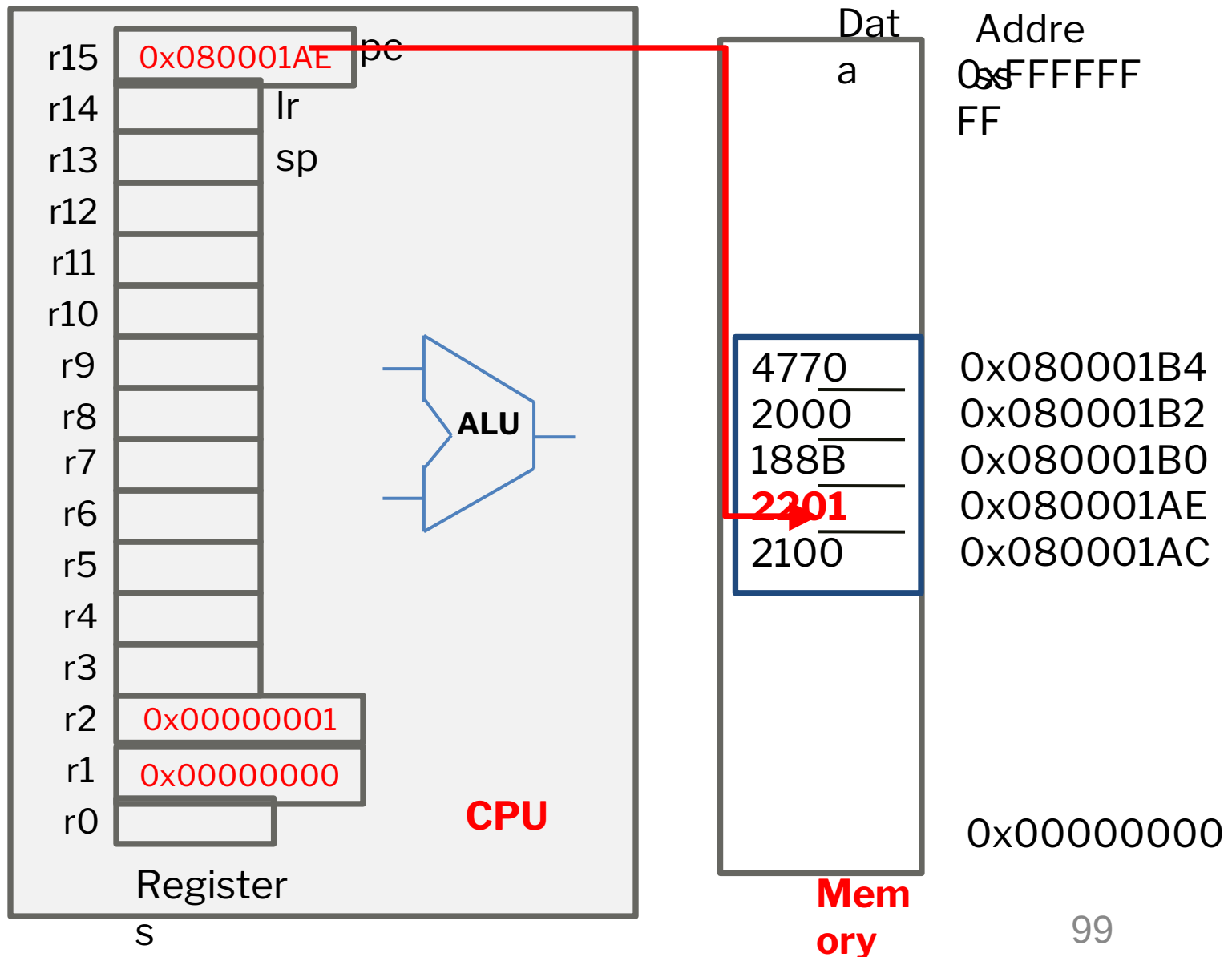


Execute Instruction: **MOVS r1, #0x00**



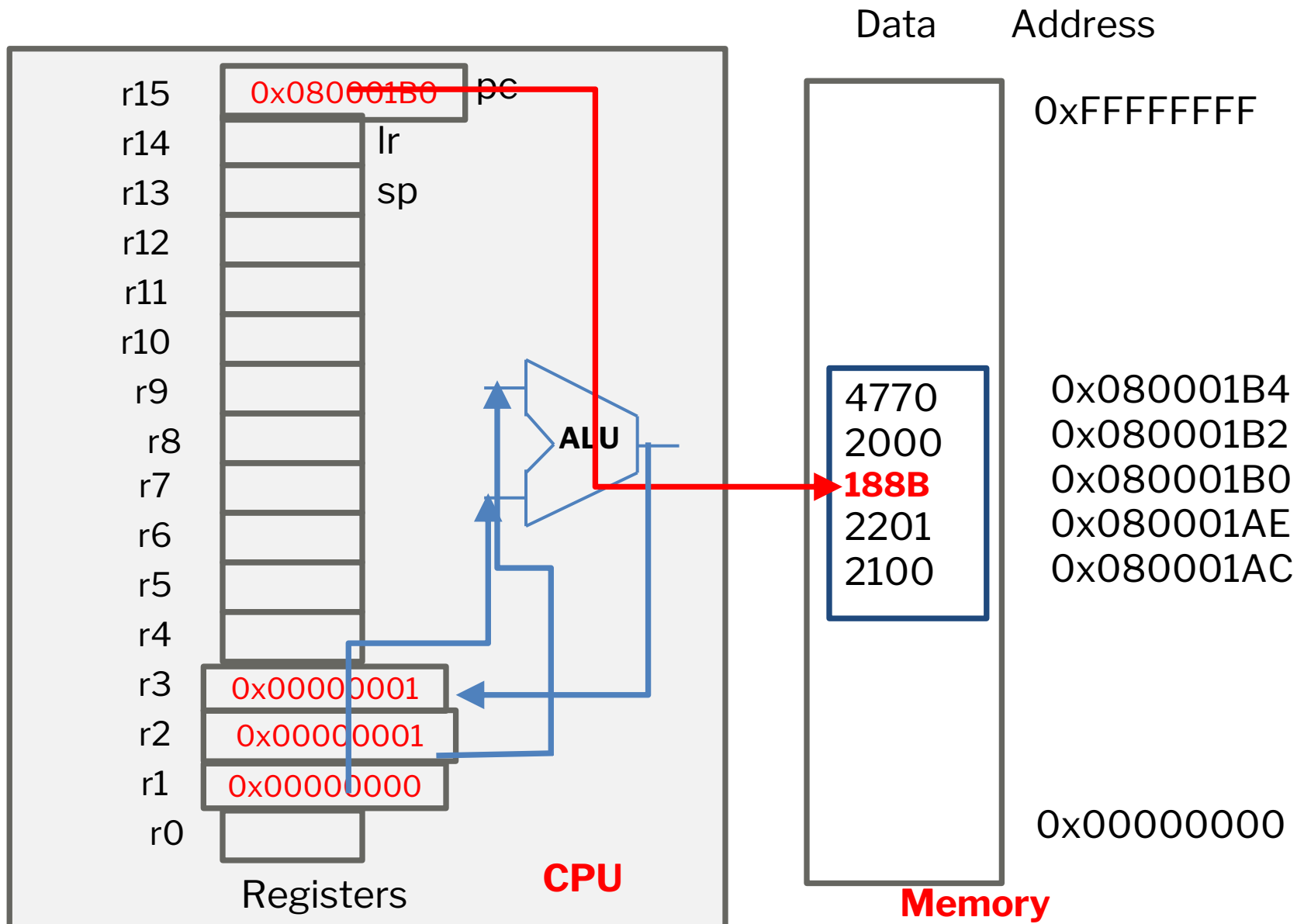
Fetch Next Instruction: $pc = pc + 2$

Decode & Execute: 2201 = **MOVS r2, #0x01**



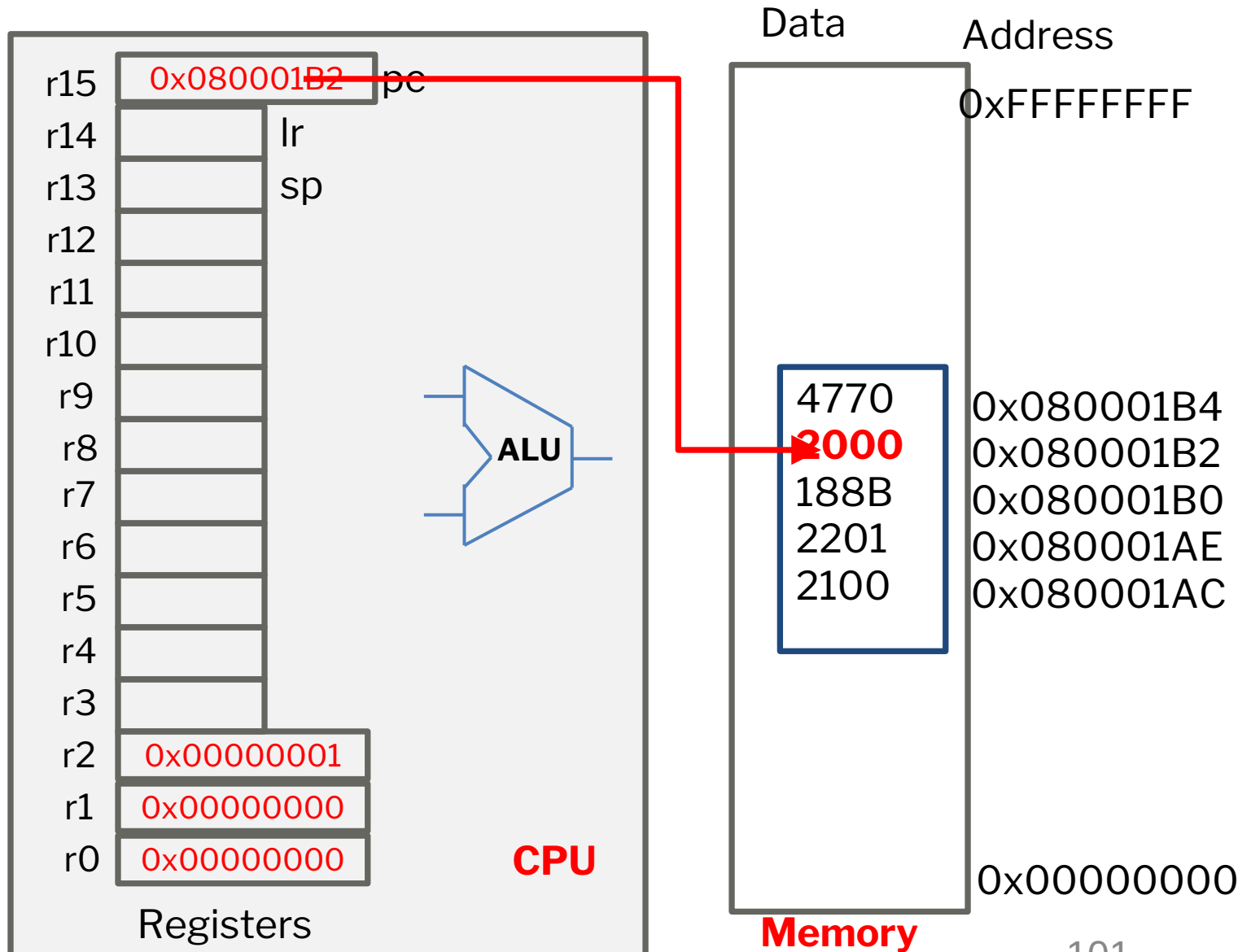
Fetch Next Instruction: $pc = pc + 2$

Decode & Execute: $188B = \text{ADDS } r3, r1, r2$



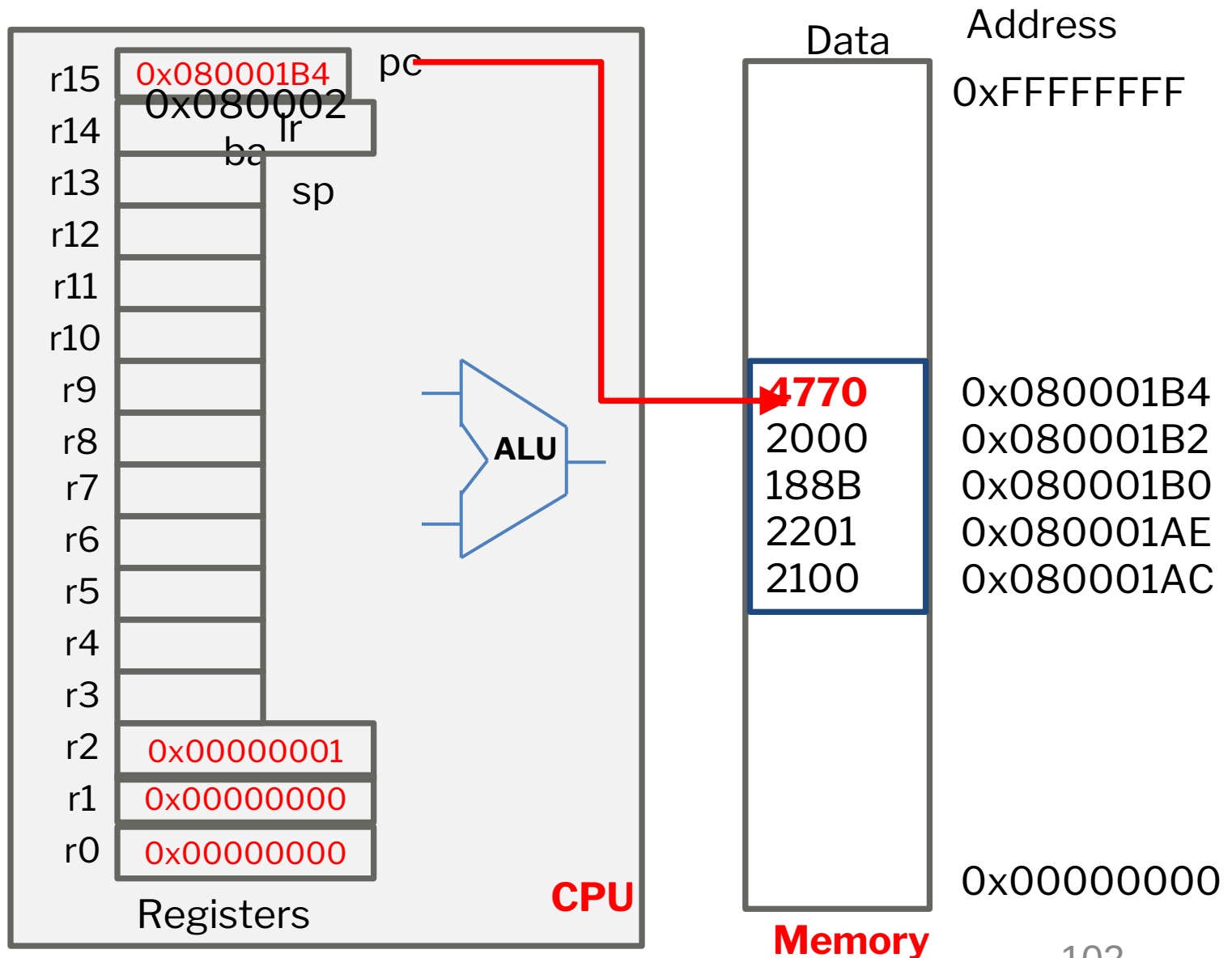
Fetch Next Instruction: $pc = pc + 2$

Decode & Execute: 2000 = **MOVS r0, #0x00**



Fetch Next Instruction: $pc = pc + 2$

Decode & Decode: 4770 = **BX Ir**

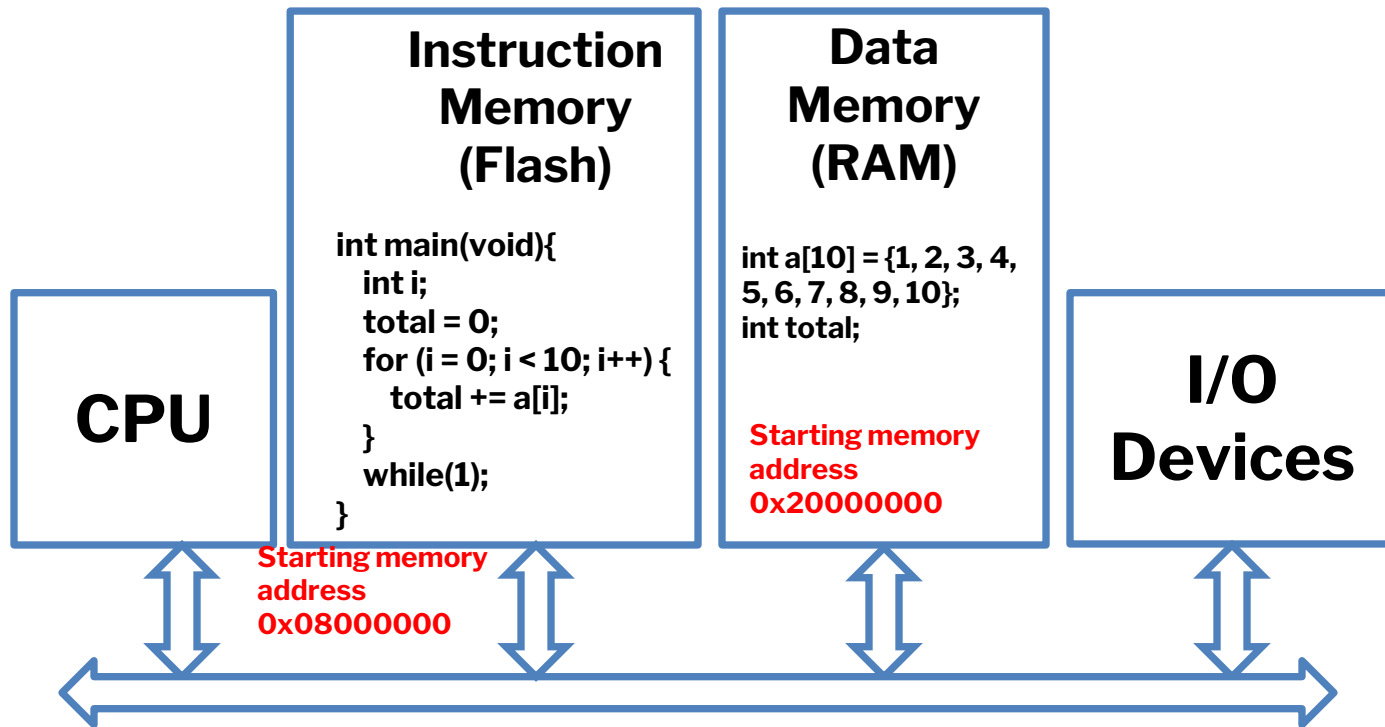


Example: Calculate the Sum of an Array

```
int a[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
int total;

int main(void){
    int i;
    total = 0;
    for (i = 0; i < 10; i++) {
        total += a[i];
    }
    while(1);
}
```

Example: Calculate the Sum of an Array



Example: Calculate the Sum of an Array

Instruction Memory (Flash)

```
int main(void){
    int i;
    total = 0;
    for (i = 0; i < 10;
    i++){
        total += a[i];
    }
    while(1);
}
```

Starting memory
address
0x08000000

```
0010 0001 0000
0000
0100 1010 0000
1000
0110 0000 0001
0001
0010 0000 0000
0000
1110 0000 0000
1000
0100 1001 0000
0111
1111 1000 0101
0001
0001 0000 0010
0000
0100 1010 0000
0100
0110 1000 0001
0010
0100 0100 0001
0001
0100 1010 0000
0011
0110 0000 0001
0001
```

```
MOVS r1, #0x00
LDR r2, = total_addr
STR r1, [r2, #0x00]
MOVS r0, #0x00
B Check
Loop: LDR r1, = a_addr
LDR r1, [r1, r0, LSL #2]
LDR r2, = total_addr
LDR r2, [r2, #0x00]
ADD r1, r1, r2
LDR r2, = total_addr
STR r1, [r2, #0x00]
ADDS r0, r0, #1
Check: CMP r0, #0x0A
BLT Loop
NOP
Self: B Self
```

Example: Calculate the Sum of an Array

Data Memory (RAM)

```
int a[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};  
int total;
```

Assume the starting memory address of
the data memory is 0x20000000

0x20000000	0x00000001
0x20000004	0x00000002
0x20000008	0x00000003
0x2000000C	0x00000004
0x20000010	0x00000005
0x20000018	0x00000006
0x2000001C	0x00000007
0x20000020	0x00000008
0x20000024	0x00000009
0x20000028	0x0000000A
0x2000002C	0x00000000
0x20000030	0x00000000

Memory
address
in words

Memory
content

a[0] = 0x00000001

a[1] = 0x00000002

a[2] = 0x00000003

a[3] = 0x00000004

a[4] = 0x00000005

a[5] = 0x00000006

a[6] = 0x00000007

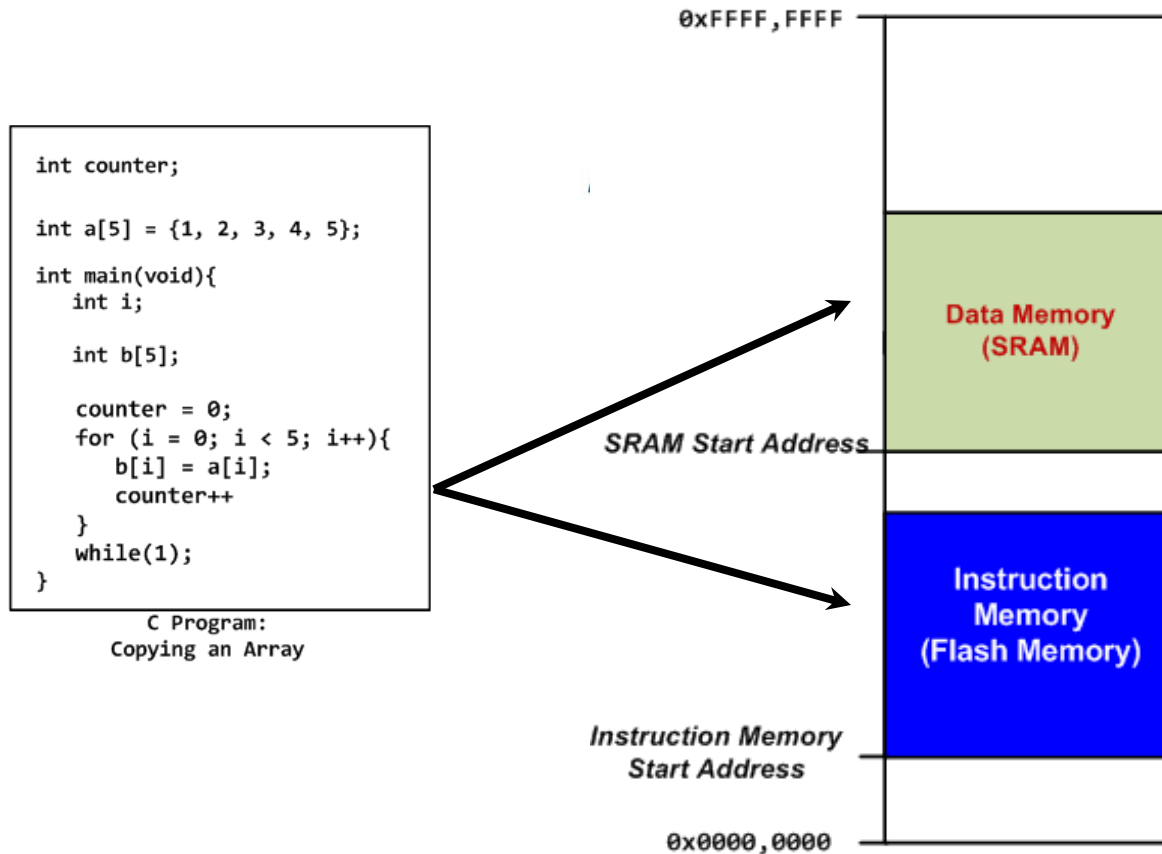
a[7] = 0x00000008

a[8] = 0x00000009

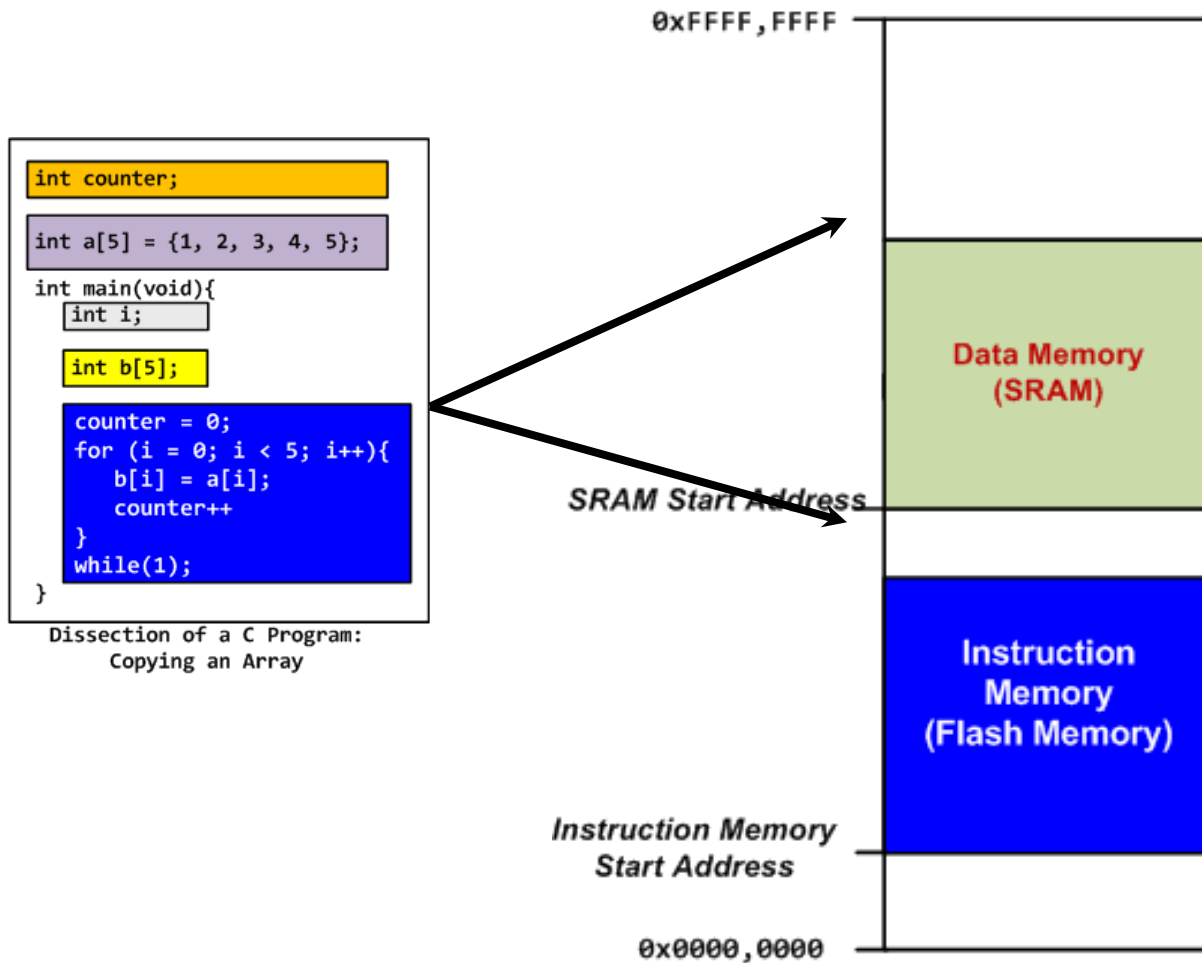
a[9] = 0x0000000A

total = 0x00000000

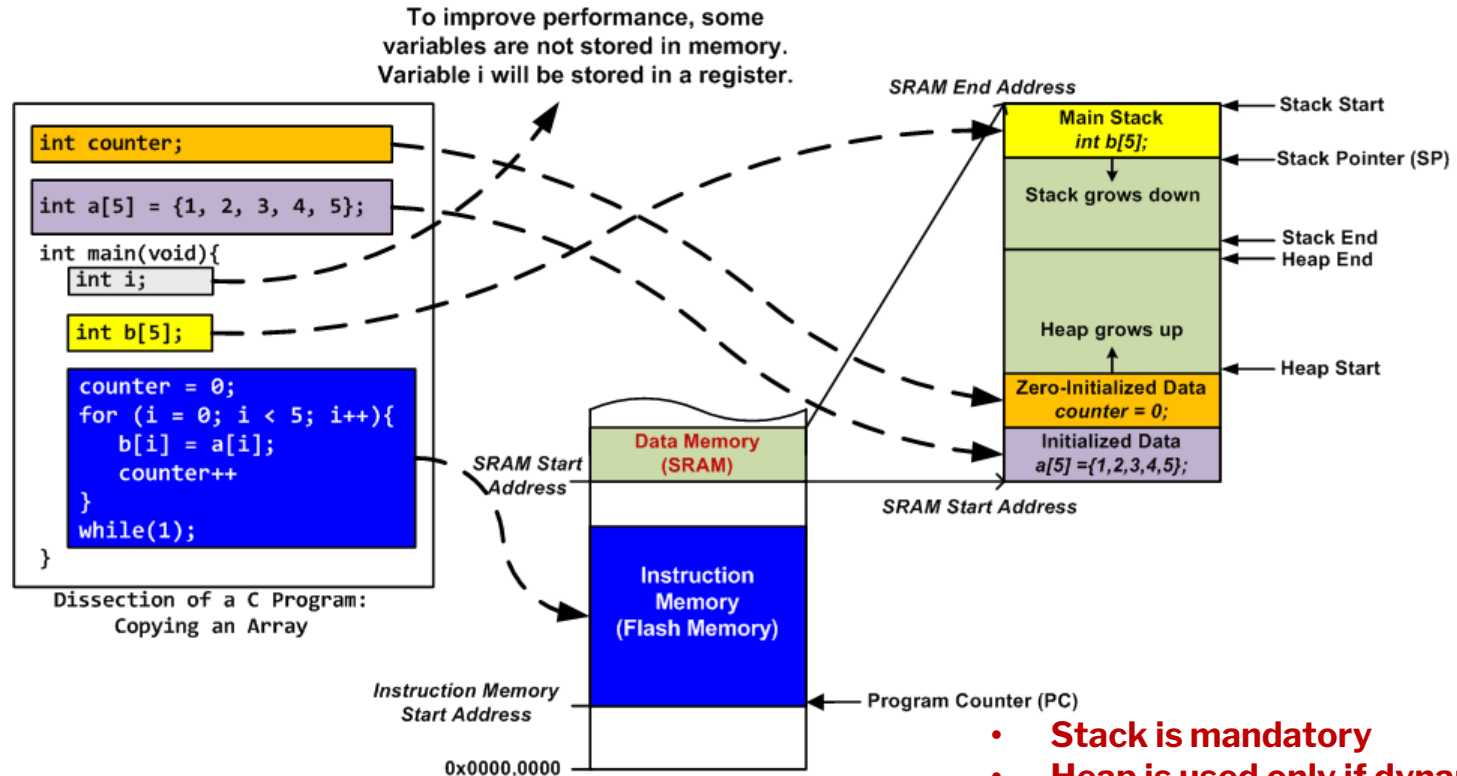
Loading Code and Data into Memory



Loading Code and Data into Memory



Loading Code and Data into Memory



- **Stack is mandatory**
- **Heap is used only if dynamic allocation (e.g. *malloc*, *calloc*) is used.**